GNU-Radio入门指南

SDR领域著名品牌:

NATIONAL INSTRUMENTS

Ettus Research™
A National Instruments Brand

威视锐®
V3 Technology,.Ltd.

课程安排及要求

《现代通信与安全综合实验》课程安排

通信基础实验

通信安全实验

授课方式：课堂和在线

课堂：分配试验任务并指导实验内容

在线：按时完成在线实验内容

撰写实验报告

实验报告包括:

- 封面（题目、个人信息）
- 实验任务
- 实验原理
- 实验步骤（包括参数设置）
- 实验结果（包括.grc流图）
- 问题分析（讲义中标红部分）

软件定义无线电(SDR)介绍

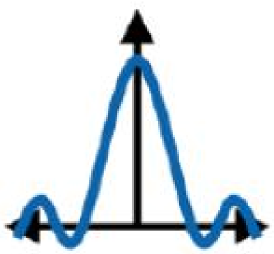
SDR设计理念

主流产品介绍

开发环境介绍

NI倡导的“图形化系统设计”理念

基于软件无线电的通信教学架构



统一平台，从理论到实现

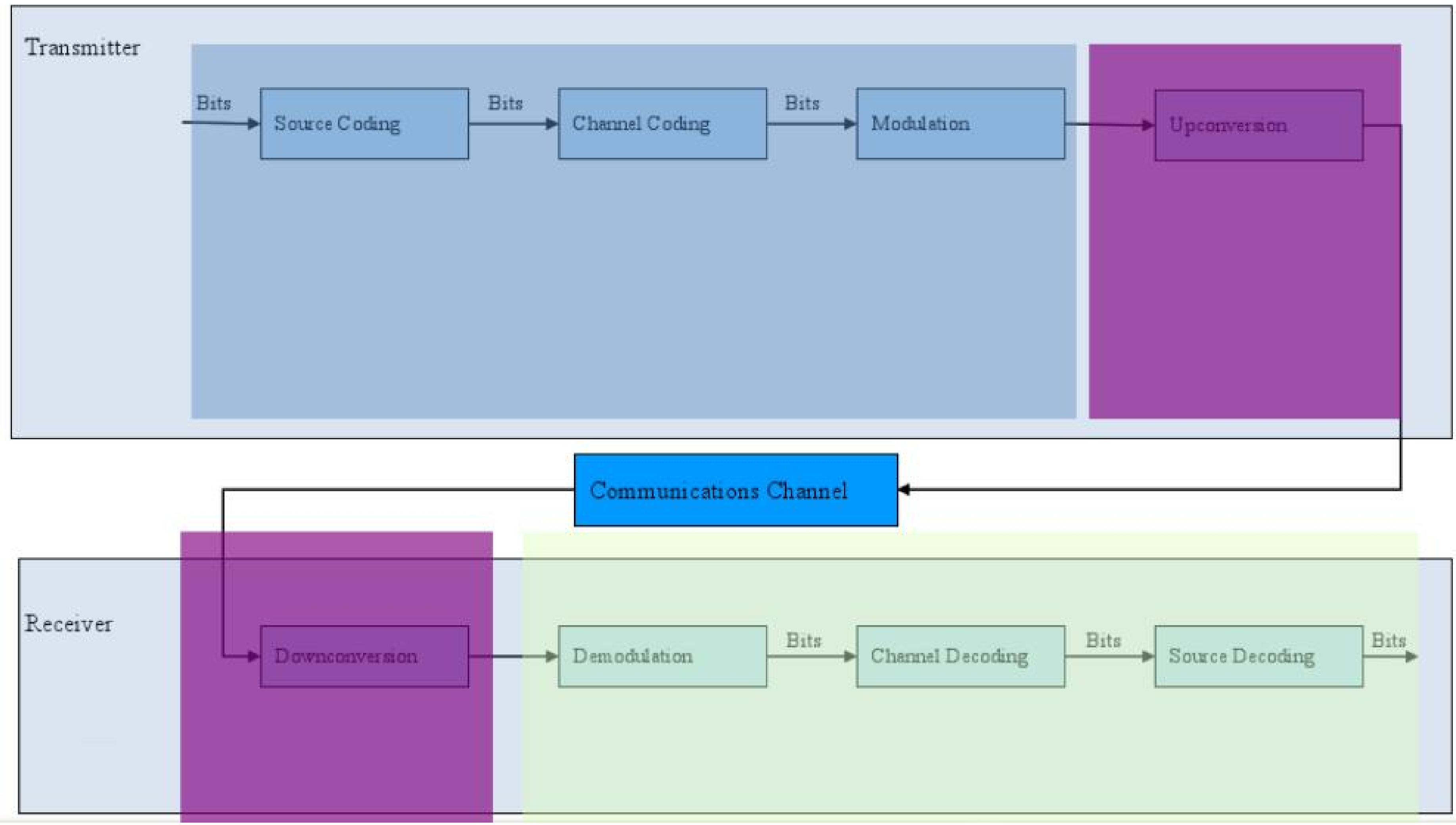
通信理论教学实践的完整流程

课堂学习	▶	课堂实验	▶	系统原型	▶	项目实现
理论		设计		原型		实现

软件无线电的定义

- 软件无线电的定义
  - **功能：** 多频段、多模式电台
  - **硬件：** 用**总线方式连接**的**标准化、模块化**的各硬件单元作为其基本平台
  - **软件：** 无线通信功能主要由**软件加载**来实现
- 工业界主流的通信系统设计理念
  - 通过运行不同的软件算法，可实时配置其通信功能，从而提供不同的无线通信业务。

真实通信系统的构建流程



完整的通信系统设计软件

调制解调

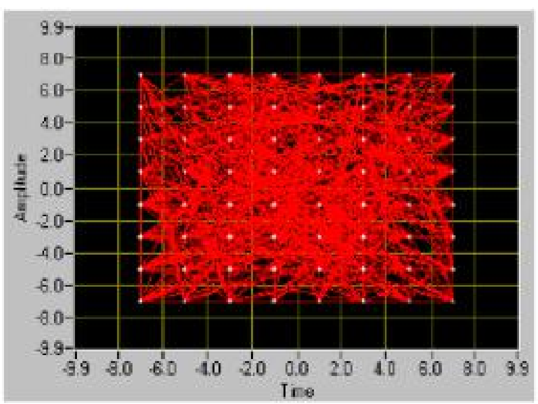
- 4, 16, 64, 256 QAM
- BPSK, QPSK, 16-PSK
- ASK, FSK, MSK, GMSK
- AM, FM, PM
- OFDM
- EVM, MER
- Quadrature Skew
- IQ Imbalance/DC Offset
- Frequency Offset
- Rho
- Bit Error Rate Testing

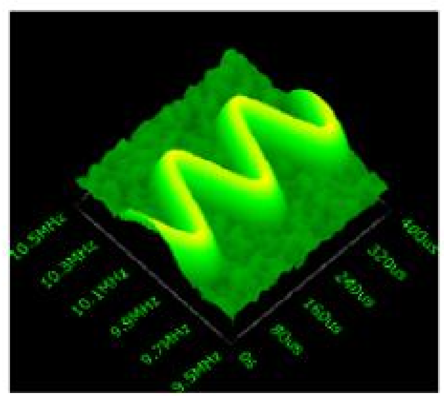
频谱测量

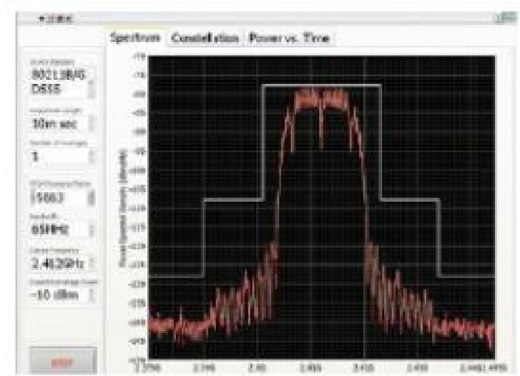
- THD, SINAD, SNR, SFDR
- Specific harmonic level
- Power in band
- Adjacent channel power
- Occupied bandwidth
- Peak detection
- Zoom FFT
- Frequency Response
- Joint Time Frequency Analysis …

通信协议

- GSM/EDGE
- WCDMA&HSPA+
- LTE
- Wimax
- ZigBee
- RFID
- WLAN
- GPS
- AM/FM
- Bluetooth
- CDMA2000
- TD-SCDMA…

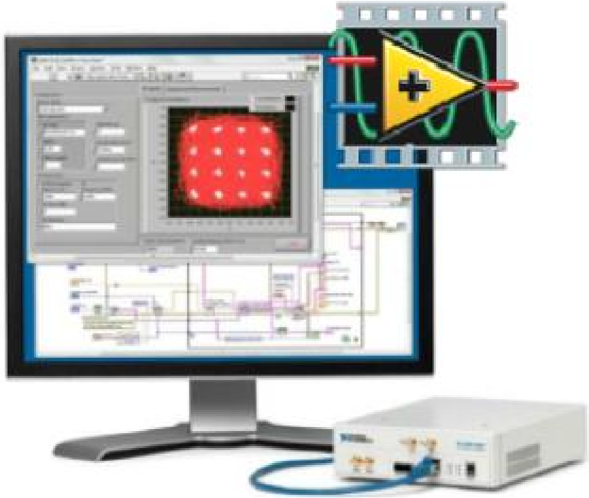


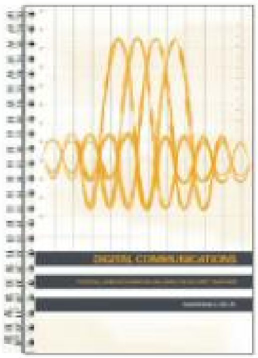




软件无线电教学平台——NI USRP

- 易上手
- 系统级
- 价格合理
- 适用于教学和科研起步



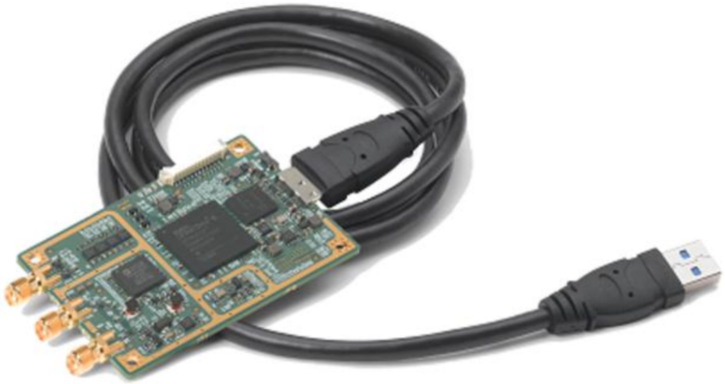


Digital Communications Lab Manual
Authored by Dr. Robert Heath

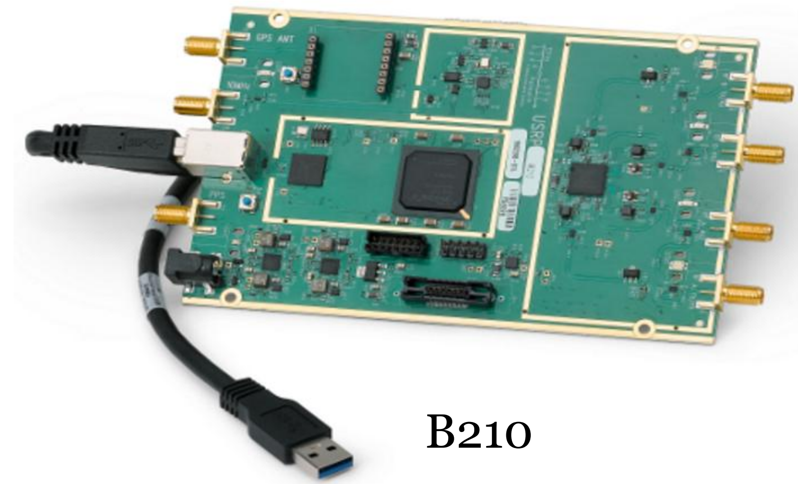

Ettus Research
USRP N210


NATIONAL INSTRUMENTS
RX 1
TX 1
RX 2
NI USRP-2920
50 MHz - 2.2 GHz
REF IN
PPS IN
MIMO EXPANSION
GB ETHERNET
A
B
C
D
E
F
POWER


USRP
security.tencent.com



B200

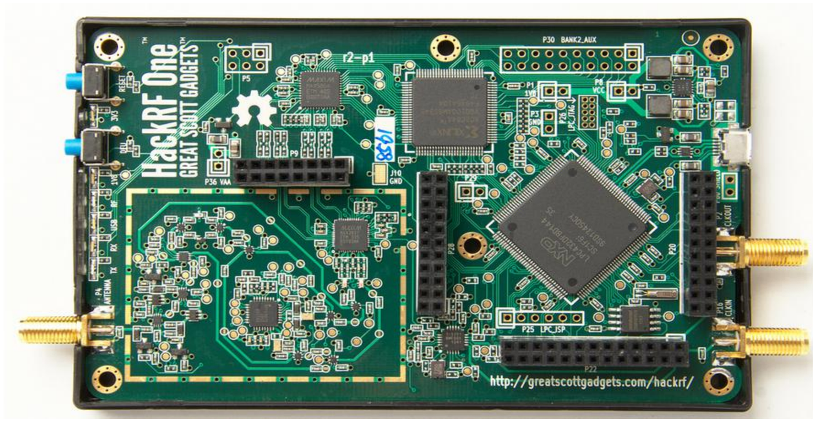


B210

B系列的主要特点

- 高度集成，射频范围从70 MHz–6 GHz；
- 支持最大实时带宽56MHz，61.44MS/s的基准采样率；
- 支持高速的USB 3.0连接；
- 开源的UHD支持多种框架；
- 用户可编写的Spartan 6 XC6SLX150 FPGA；
- 为模拟设备在AD9361射频早期访问原型平台，一个完全集成的混合信号基带直接转换收发器。

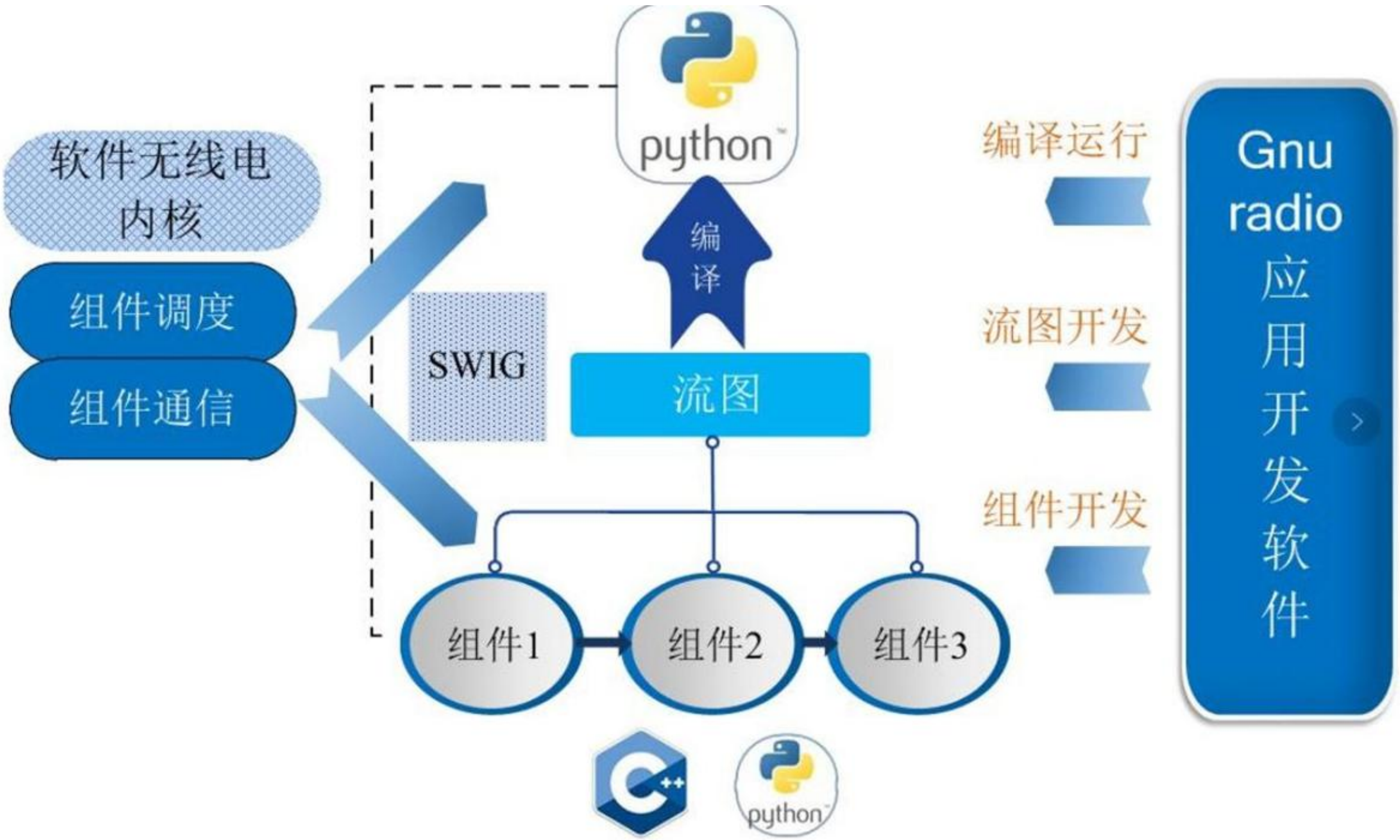
HackRF One：一款开源的软件定义无线电设备



	HackRF	bladeRF		USRP		
		x40	x115	B100 Starter	B200	B210
Radio Spectrum	30 MHz – 6 GHz	300 MHz – 3.8 GHz		50 MHz – 2.2 GHz [1]	50MHz – 6 GHz	
Bandwidth	20 MHz	28 MHz		16 MHz [2]	61.44 MHz [3]	
Duplex	Half	Full		Full	Full	2x2 MIMO
Sample Size (ADC/DAC)	8 bit	12 bit		12 bit / 14 bit	12 bit	
Sample Rate (ADC/DAC)	20 Msps	40 Msps		64 Msps / 128 Msps	61.44 Msps	
Interface (Speed)	USB 2 HS (480 megabit)	USB 3 (5 gigabit)		USB 2 HS (480 megabit)	USB 3 (5 gigabit)	
FPGA Logic Elements	[4]	40k	115k	25k	75k	150k
Microcontroller	LPC43XX	Cypress FX3		Cypress FX2	Cypress FX3	
Open Source	Everything	HDL + Code Schematics		HDL + Code Schematics	Host Code [5]	
Availability	January 2014	Now		Now	Now	
Cost	$300 [6]	$420	$650	$675	$675	$1100

□USRP(Universal Software Radio Peripheral)介绍

USRP是一个灵活的**软件无线电**平台**外设**，可以与很多软件配套使用，比如：GNURadio、Matlab/Simulink、Labview等。只需要通过一根网线连通USRP与PC，就可以实现一个软件无线电系统。


软件无线电内核
组件调度
组件通信
SWIG
python
编译
流图
组件1
组件2
组件3
python
编译运行
流图开发
组件开发
Gnu radio 应用开发软件

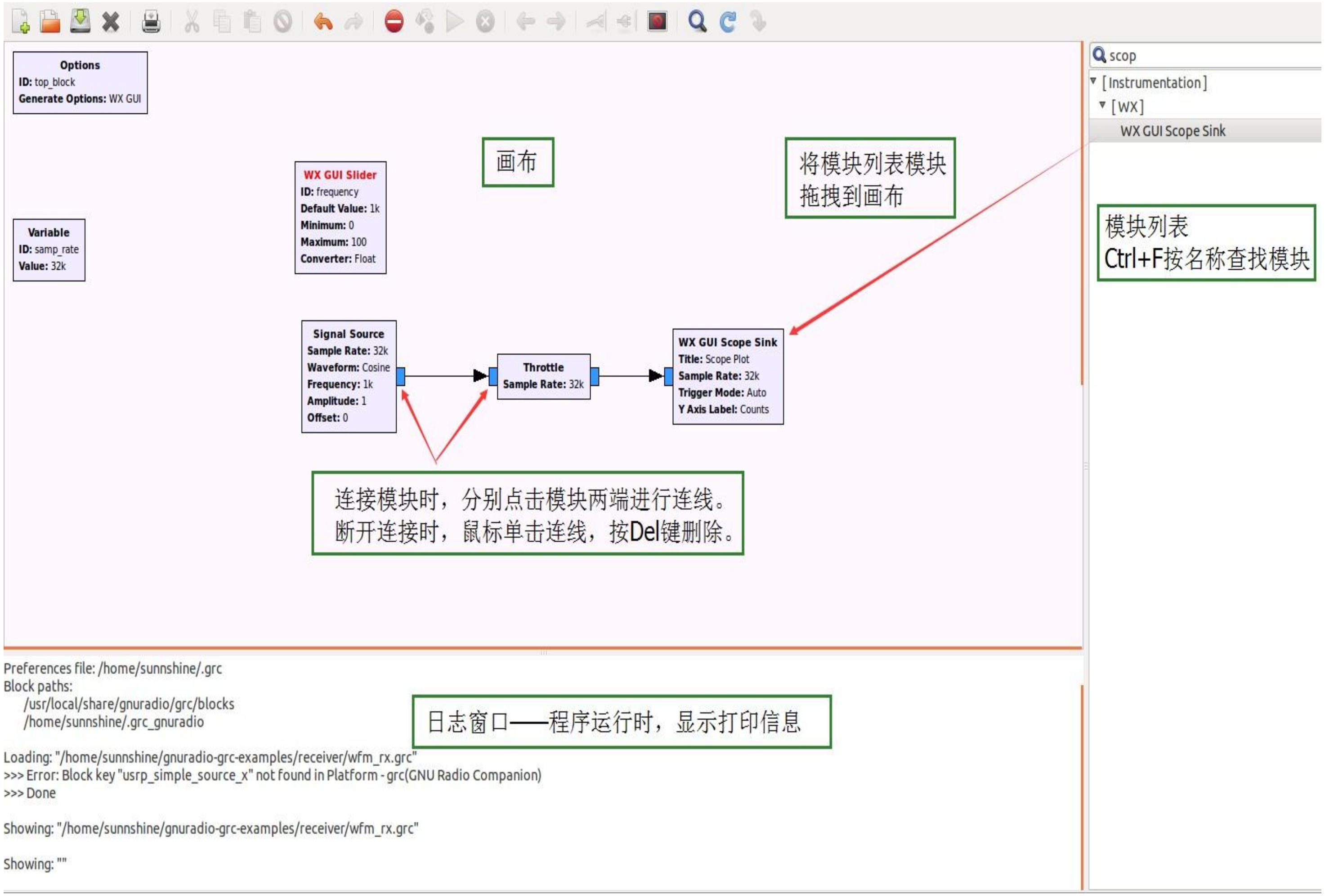
□ GNU-Radio基础

打开GNURadio Companion；

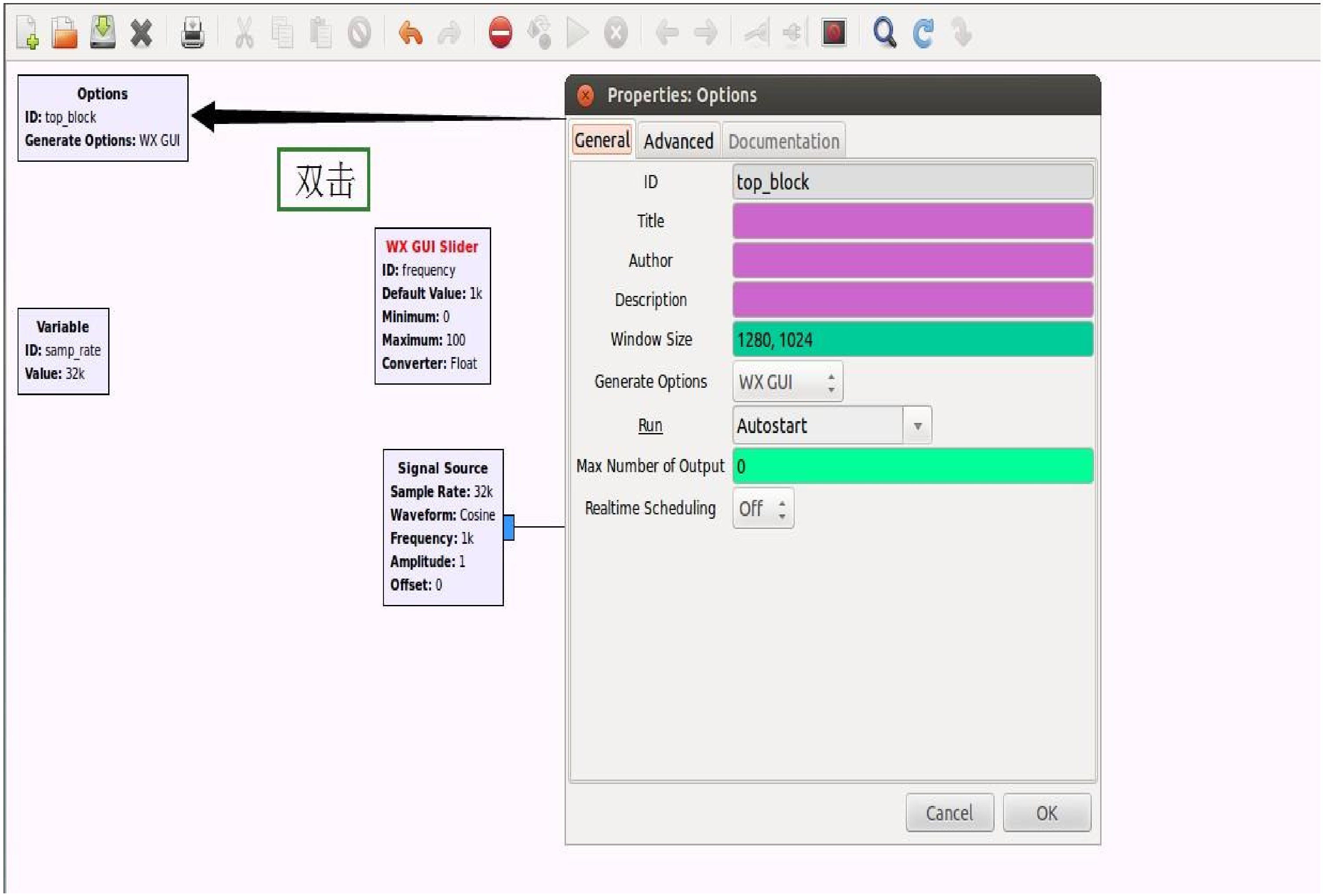
——打开终端/控制台/命令提示符

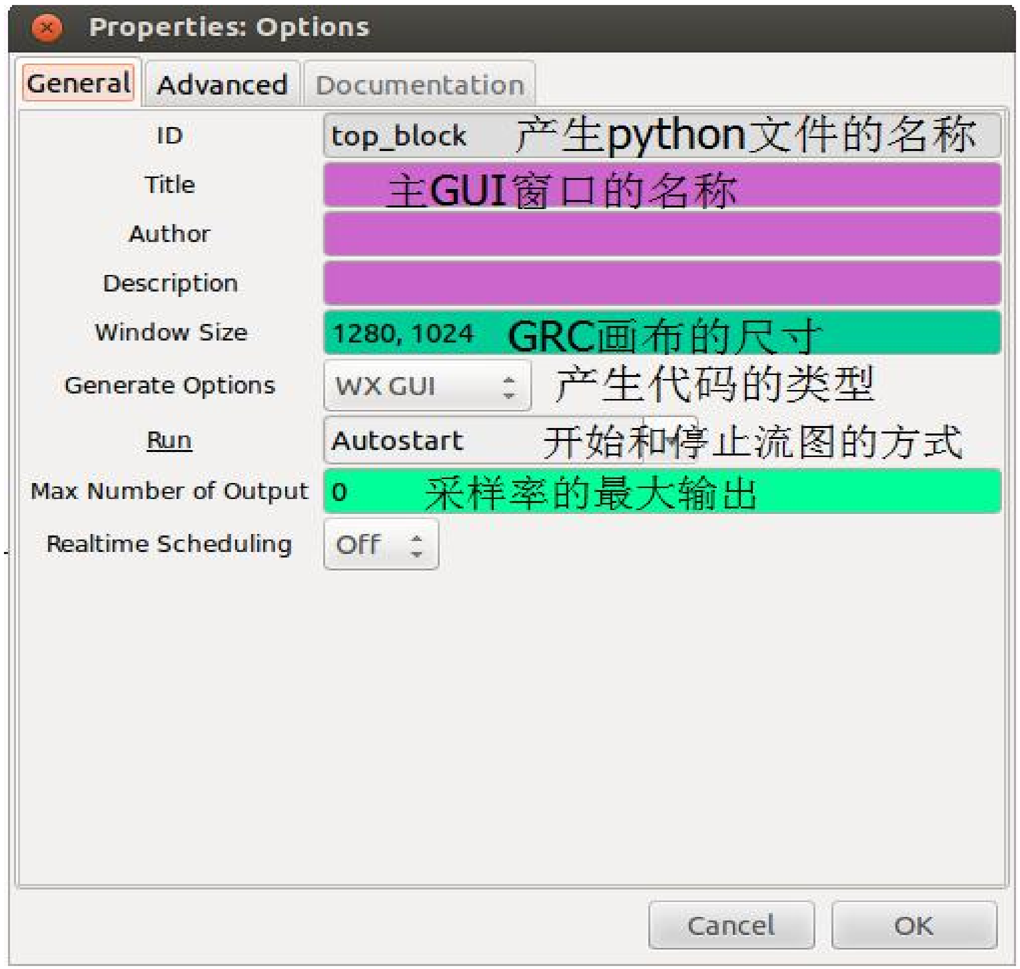
——运行‘gnuradio-companion’

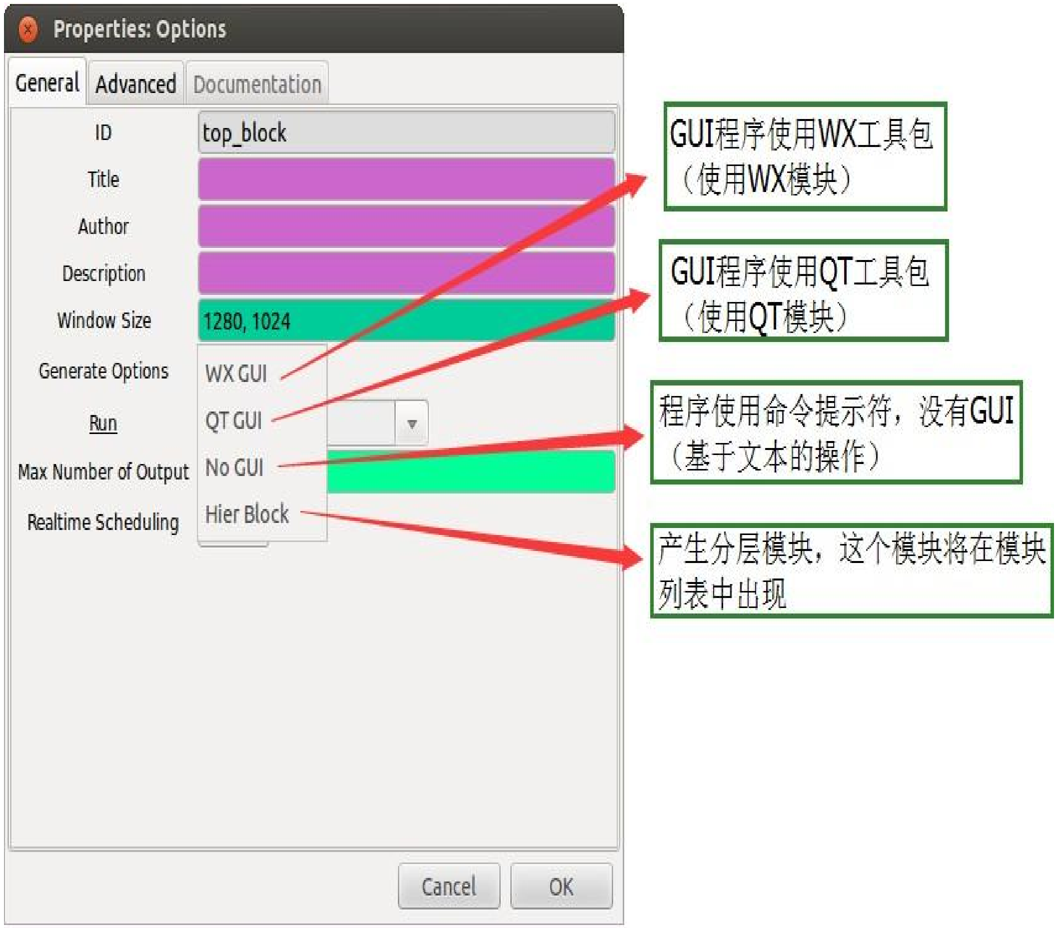
Lab 1：熟悉基本模块

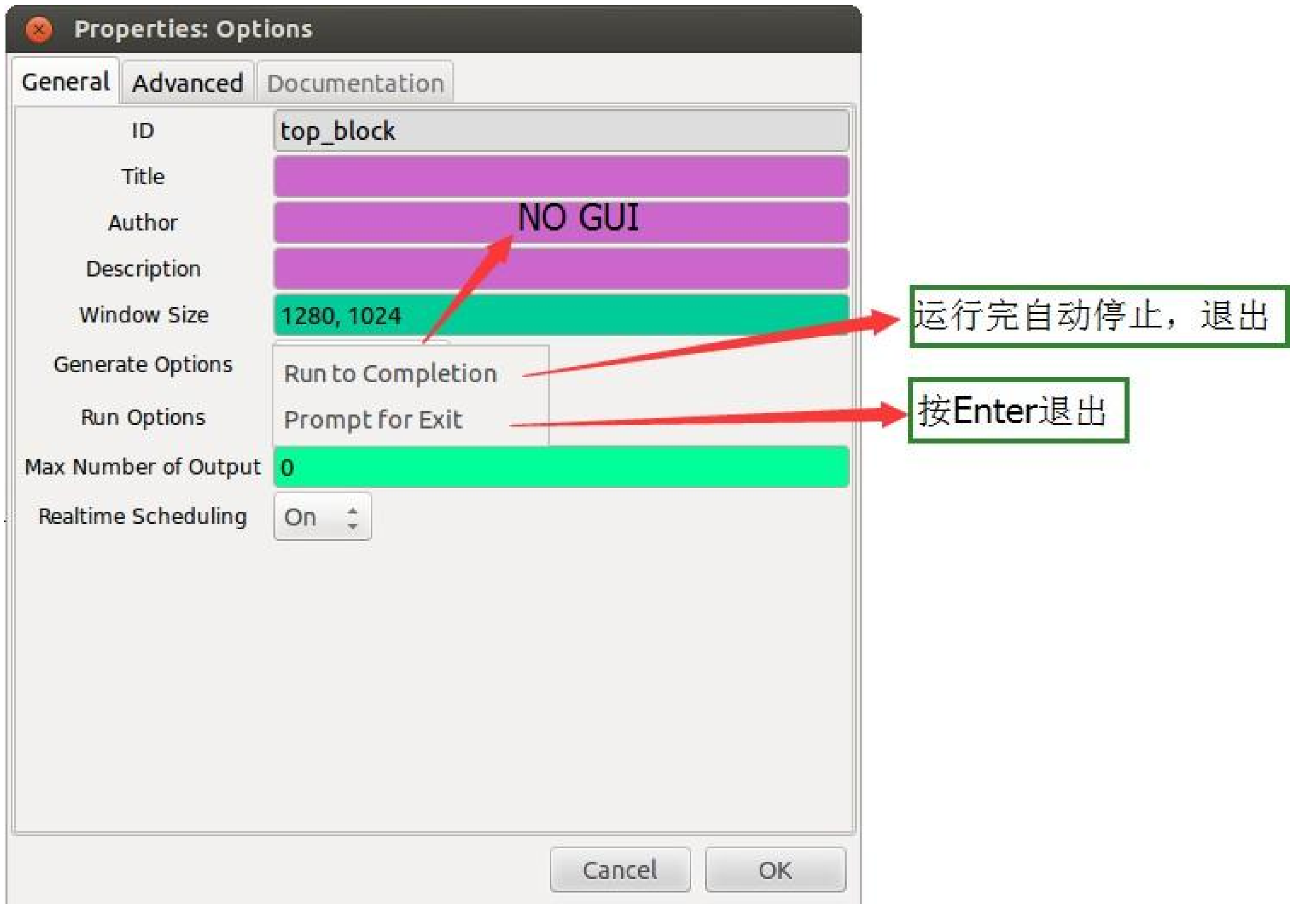


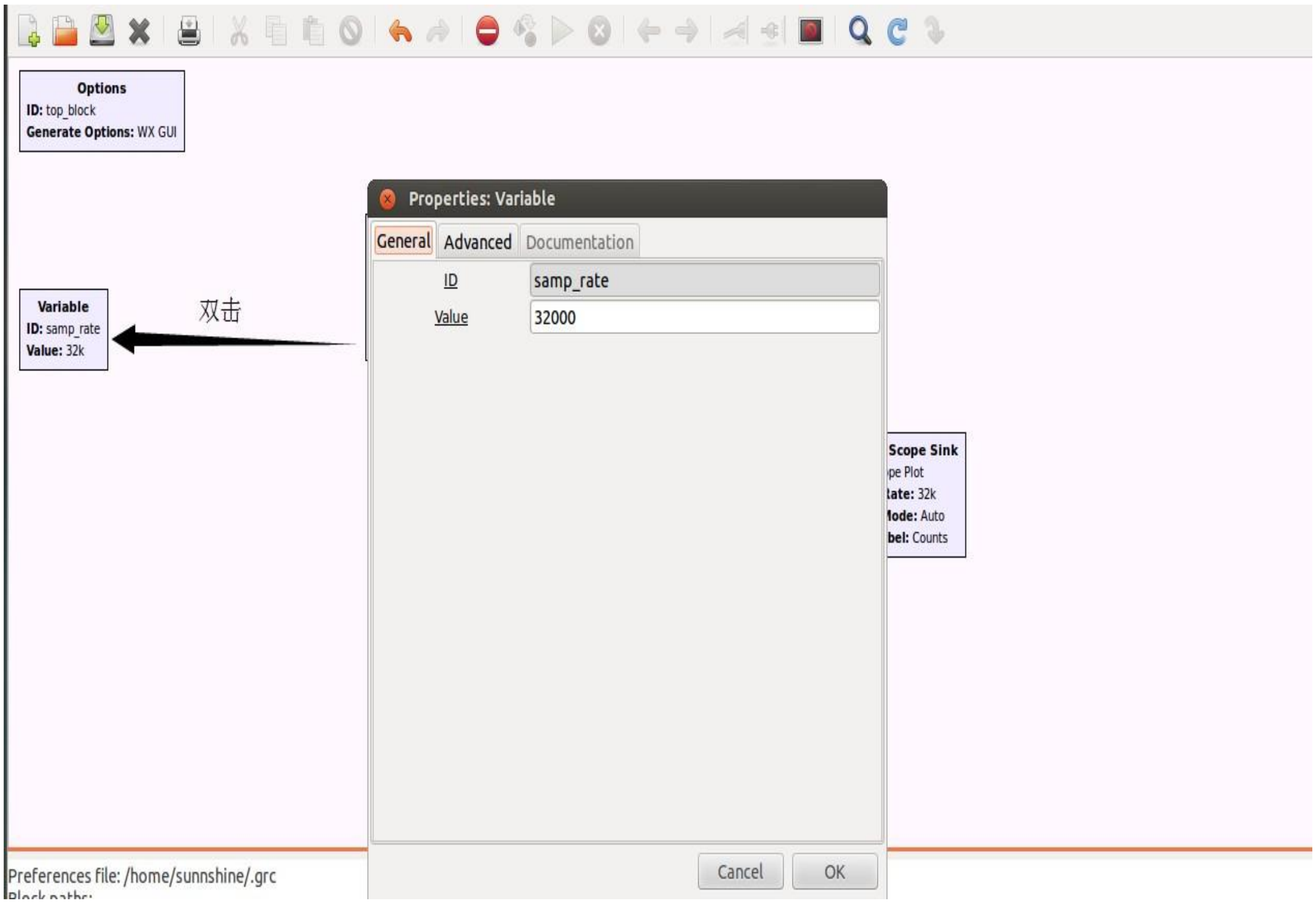
Options
ID: top_block
Generate Options: WX GUI
Variable
ID: samp_rate
Value: 32k
WX GUI Slider
ID: frequency
Default Value: 1k
Minimum: 0
Maximum: 100
Converter: Float
画布
将模块列表模块拖拽到画布
scop
[Instrumentation]
[WX]
WX GUI Scope Sink
模块列表
Ctrl+F按名称查找模块
Signal Source
Sample Rate: 32k
Waveform: Cosine
Frequency: 1k
Amplitude: 1
Offset: 0
Throttle
Sample Rate: 32k
WX GUI Scope Sink
Title: Scope Plot
Sample Rate: 32k
Trigger Mode: Auto
Y Axis Label: Counts
连接模块时，分别点击模块两端进行连线。
断开连接时，鼠标单击连线，按Del键删除。
Preferences file: /home/sunnshine/.grc
Block paths:
/usr/local/share/gnuradio/grc/blocks
/home/sunnshine/.grc_gnuradio
日志窗口——程序运行时，显示打印信息
Loading: "/home/sunnshine/gnuradio-grc-examples/receiver/wfm_rx.grc"
>>> Error: Block key "usrp_simple_source_x" not found in Platform - grc(GNU Radio Companion)
>>> Done
Showing: "/home/sunnshine/gnuradio-grc-examples/receiver/wfm_rx.grc"
Showing: ""

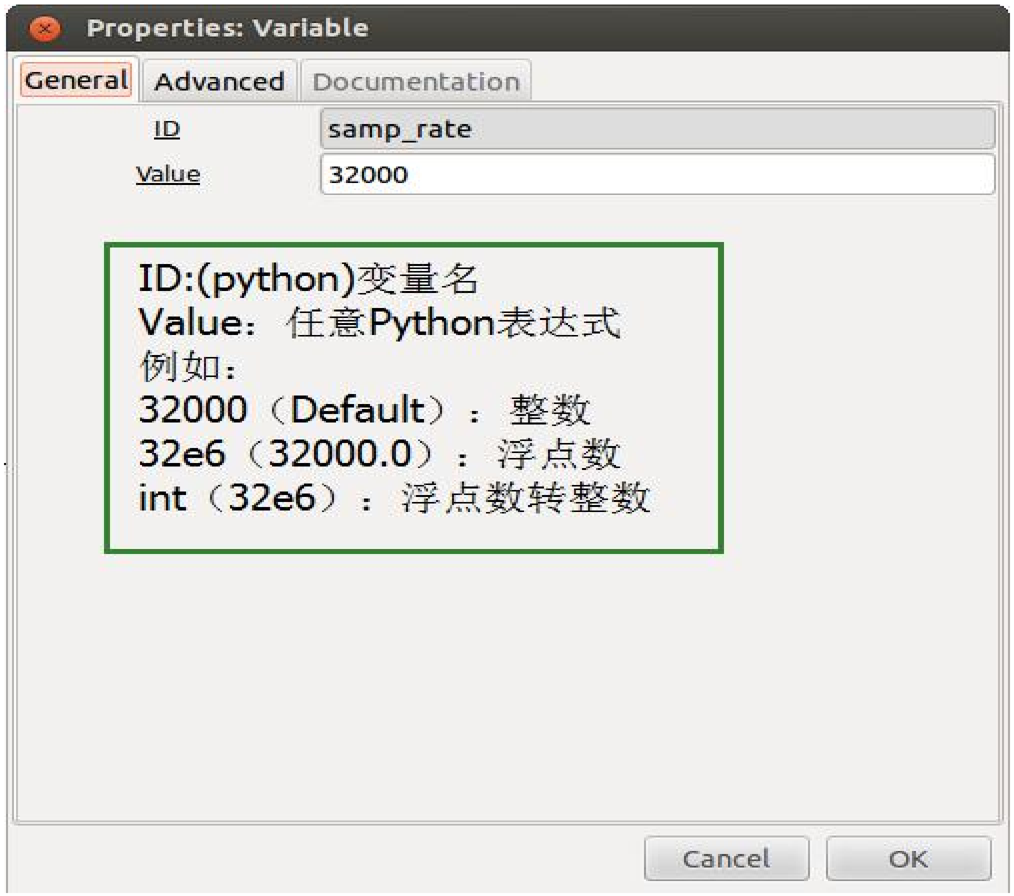

Options
ID: top_block
Generate Options: WX GUI
双击
WX GUI Slider
ID: frequency
Default Value: 1k
Minimum: 0
Maximum: 100
Converter: Float
Variable
ID: samp_rate
Value: 32k
Signal Source
Sample Rate: 32k
Waveform: Cosine
Frequency: 1k
Amplitude: 1
Offset: 0
Properties: Options
General
Advanced
Documentation
ID
top_block
Title
Author
Description
Window Size
1280, 1024
Generate Options
WX GUI
Run
Autostart
Max Number of Output
0
Realtime Scheduling
Off
Cancel
OK

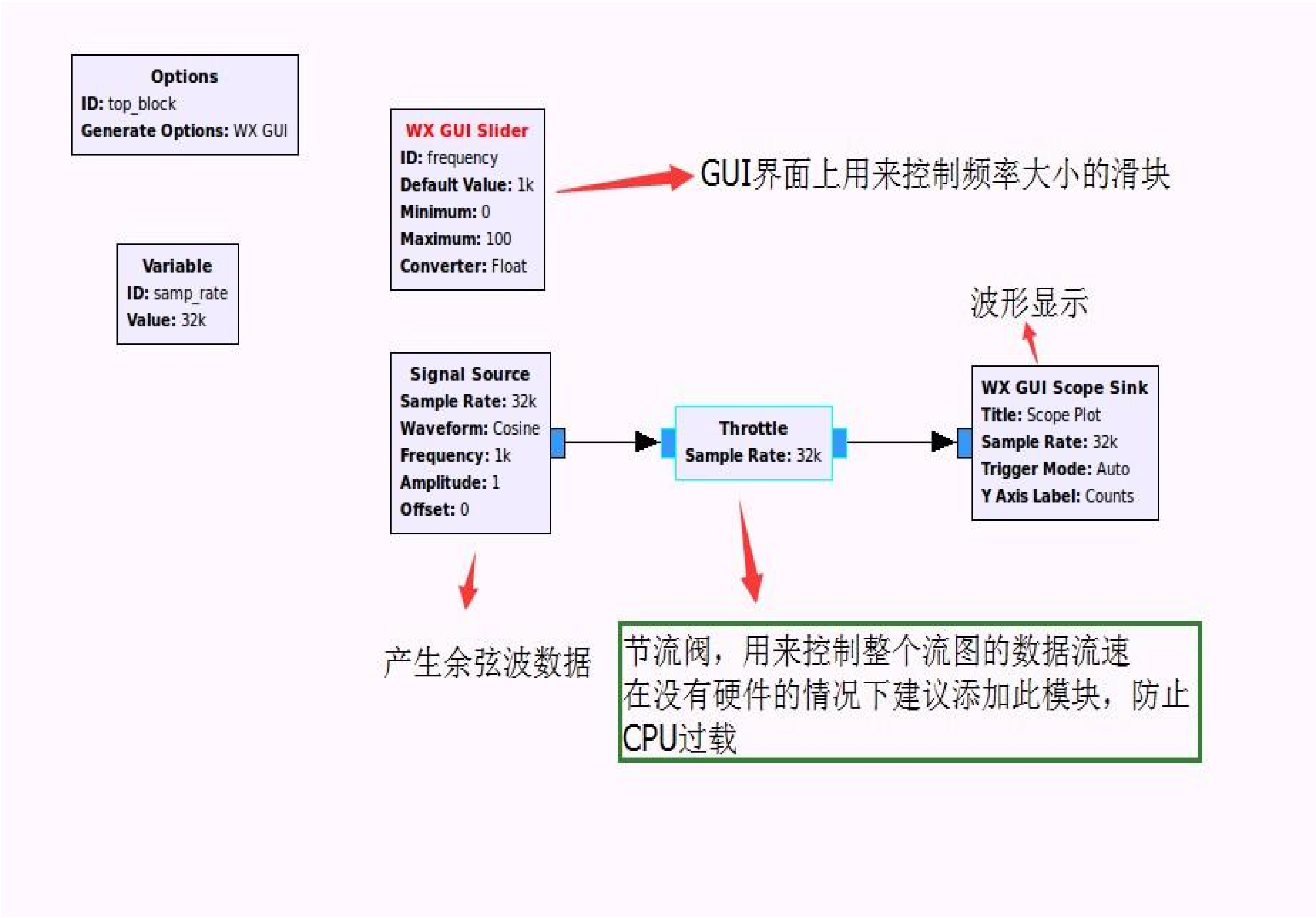

Properties: Options
General
Advanced
Documentation
ID
top_block
产生python文件的名称
Title
主GUI窗口的名称
Author
Description
Window Size
1280, 1024
GRC画布的尺寸
Generate Options
WX GUI
产生代码的类型
Run
Autostart
开始和停止流图的方式
Max Number of Output
0
采样率的最大输出
Realtime Scheduling
Off
Cancel
OK

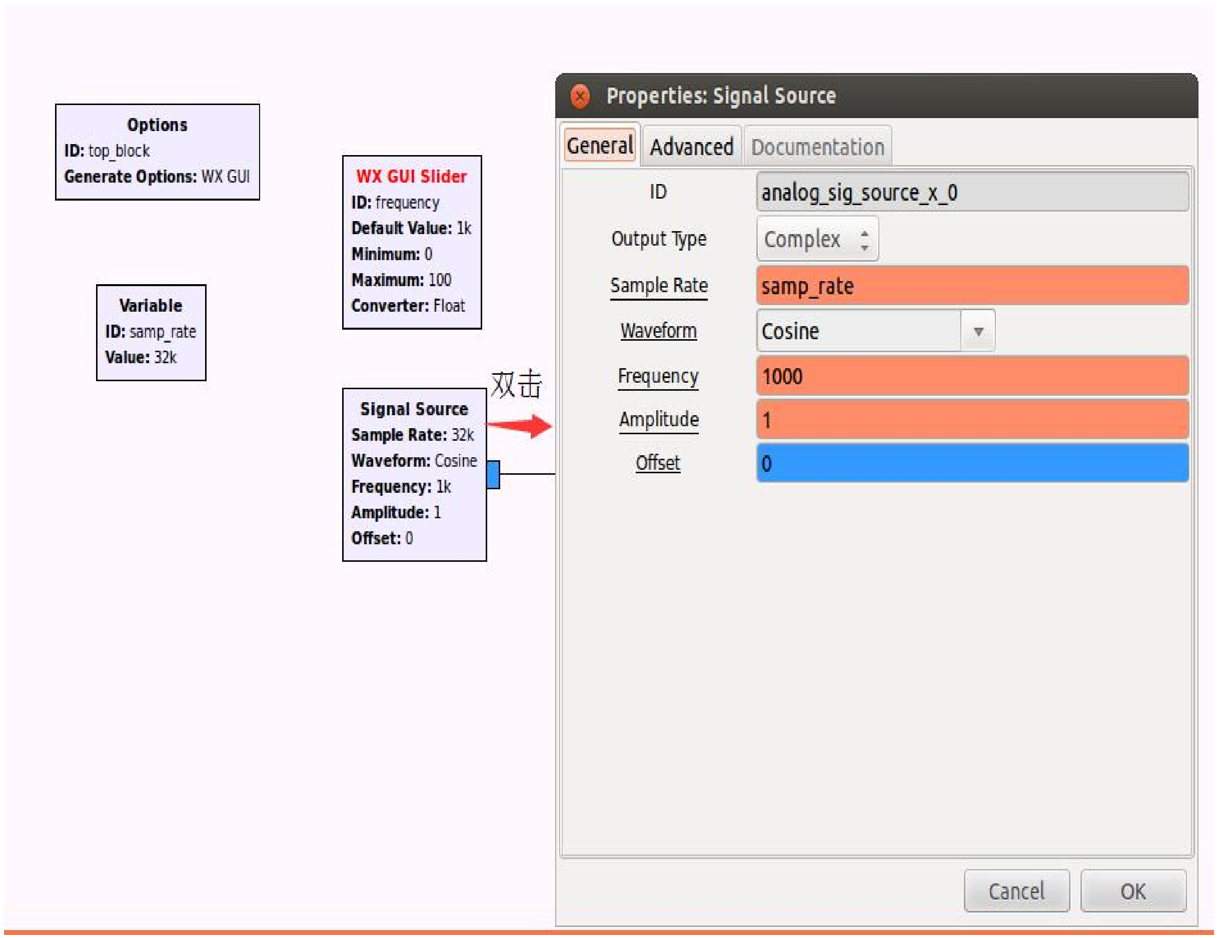

Properties: Options
General
Advanced
Documentation
ID
top_block
Title
Author
Description
Window Size
1280, 1024
Generate Options
Run
Max Number of Output
Realtime Scheduling
WX GUI
QT GUI
No GUI
Hier Block
Cancel
OK
GUI程序使用WX工具包（使用WX模块）
GUI程序使用QT工具包（使用QT模块）
程序使用命令提示符，没有GUI（基于文本的操作）
产生分层模块，这个模块将在模块列表中出现


Properties: Options
General
Advanced
Documentation
ID
top_block
Title
Author
NO GUI
Description
Window Size
1280, 1024
Generate Options
Run to Completion
Run Options
Prompt for Exit
Max Number of Output
0
Realtime Scheduling
On
Cancel
OK
运行完自动停止，退出
按Enter退出


Options
ID: top_block
Generate Options: WX GUI
Variable
ID: samp_rate
Value: 32k
双击
Properties: Variable
General
Advanced
Documentation
ID
samp_rate
Value
32000
Scope Sink
pe Plot
late: 32k
lode: Auto
bel: Counts
Cancel
OK
Preferences file: /home/sunnshine/.grc

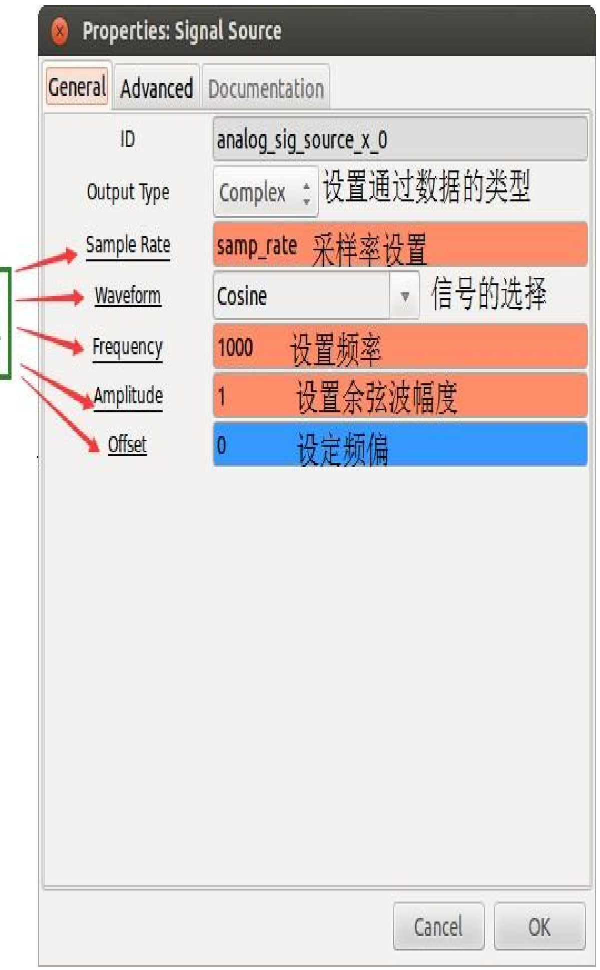

Properties: Variable
General
Advanced
Documentation
ID
samp_rate
Value
32000
ID:(python)变量名
Value：任意Python表达式
例如：
32000（Default）：整数
32e6（32000.0）：浮点数
int（32e6）：浮点数转整数
Cancel
OK

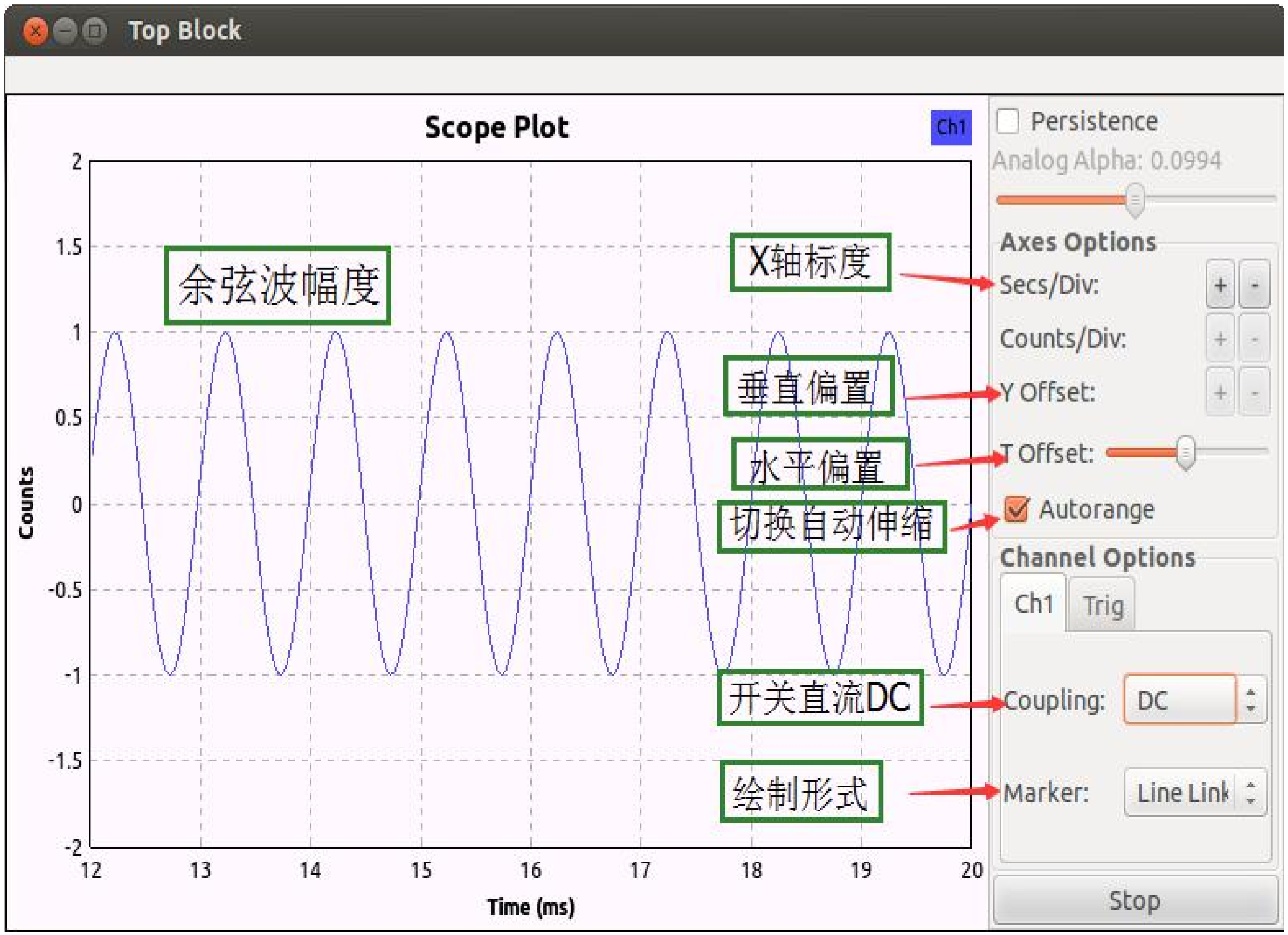

Options
ID: top_block
Generate Options: WX GUI
Variable
ID: samp_rate
Value: 32k
WX GUI Slider
ID: frequency
Default Value: 1k
Minimum: 0
Maximum: 100
Converter: Float
GUI界面上用来控制频率大小的滑块
波形显示
Signal Source
Sample Rate: 32k
Waveform: Cosine
Frequency: 1k
Amplitude: 1
Offset: 0
Throttle
Sample Rate: 32k
WX GUI Scope Sink
Title: Scope Plot
Sample Rate: 32k
Trigger Mode: Auto
Y Axis Label: Counts
产生余弦波数据
节流阀，用来控制整个流图的数据流速
在没有硬件的情况下建议添加此模块，防止
CPU过载


Options
ID: top_block
Generate Options: WX GUI
WX GUI Slider
ID: frequency
Default Value: 1k
Minimum: 0
Maximum: 100
Converter: Float
Variable
ID: samp_rate
Value: 32k
双击
Signal Source
Sample Rate: 32k
Waveform: Cosine
Frequency: 1k
Amplitude: 1
Offset: 0
Properties: Signal Source
General
Advanced
Documentation
ID
analog_sig_source_x_0
Output Type
Complex
Sample Rate
samp_rate
Waveform
Cosine
Frequency
1000
Amplitude
1
Offset
0
Cancel
OK

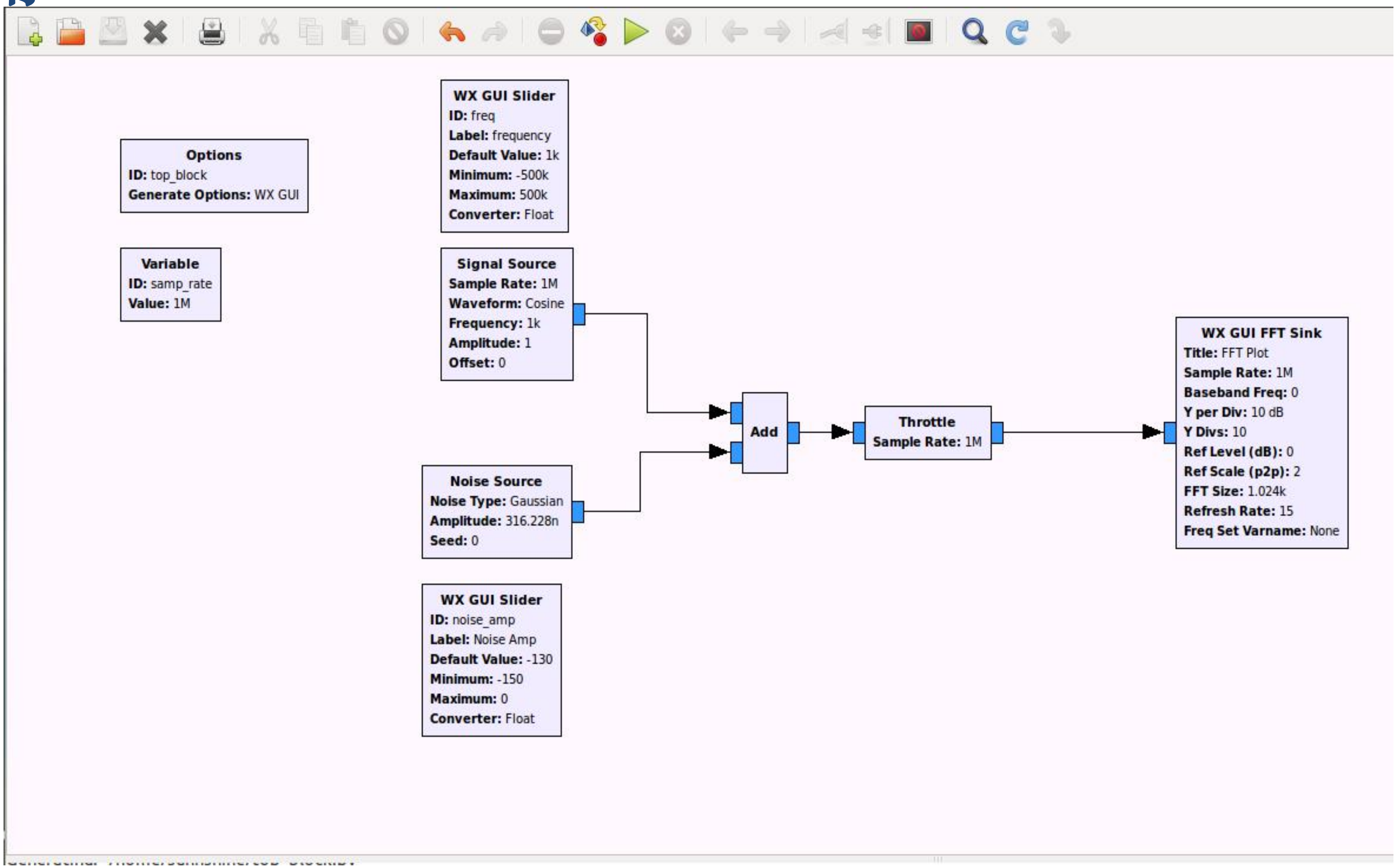
任何处理模块中的'Sample Rate'都是为DSP计算使用，而不是控制产生采样的速率。这与硬件控制采样速率有明显的区别

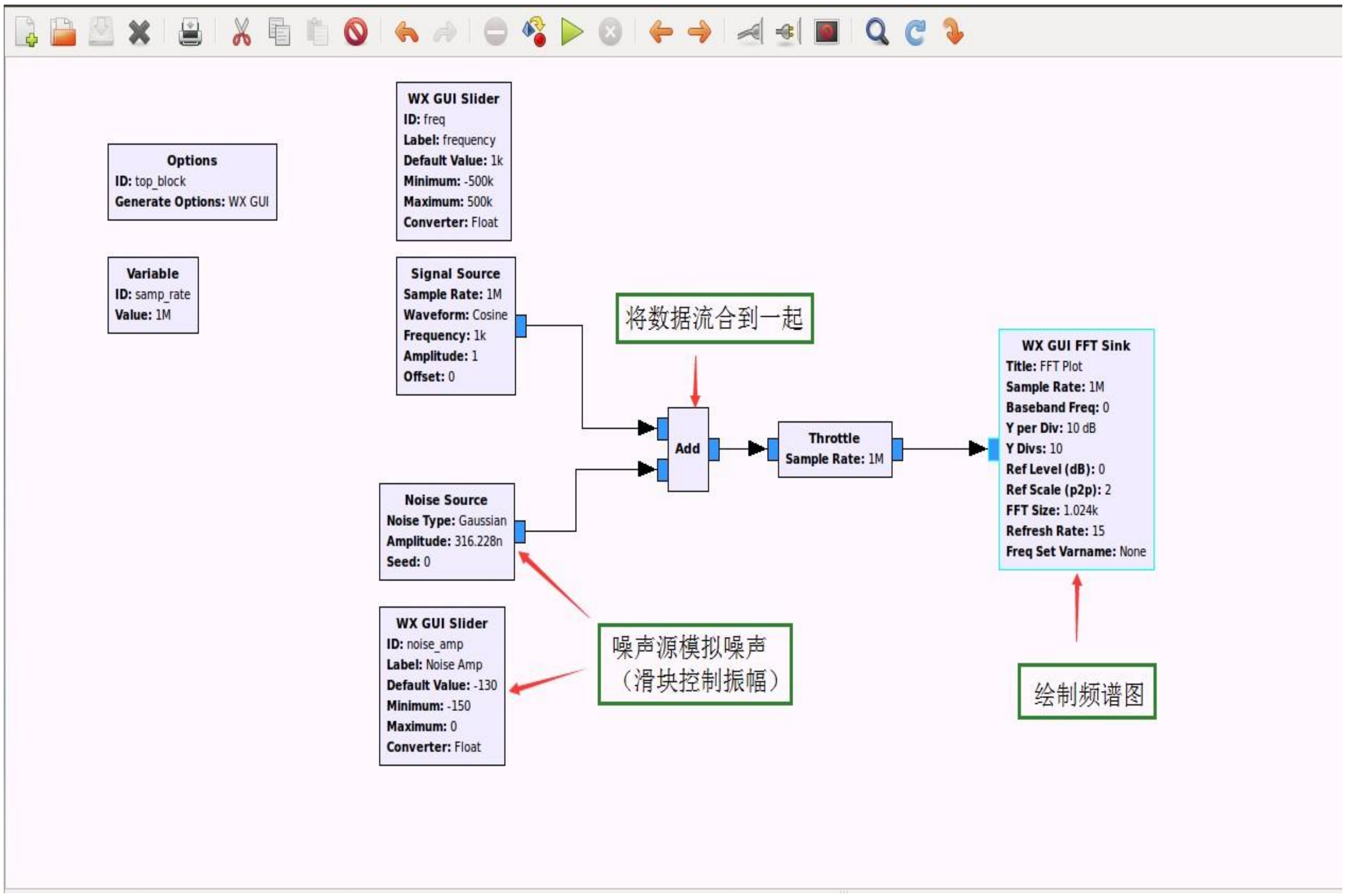
任何带下划线的参数都是可以以任意Python表达式来表述的，也可以通过变量传递参数。使用较为灵活

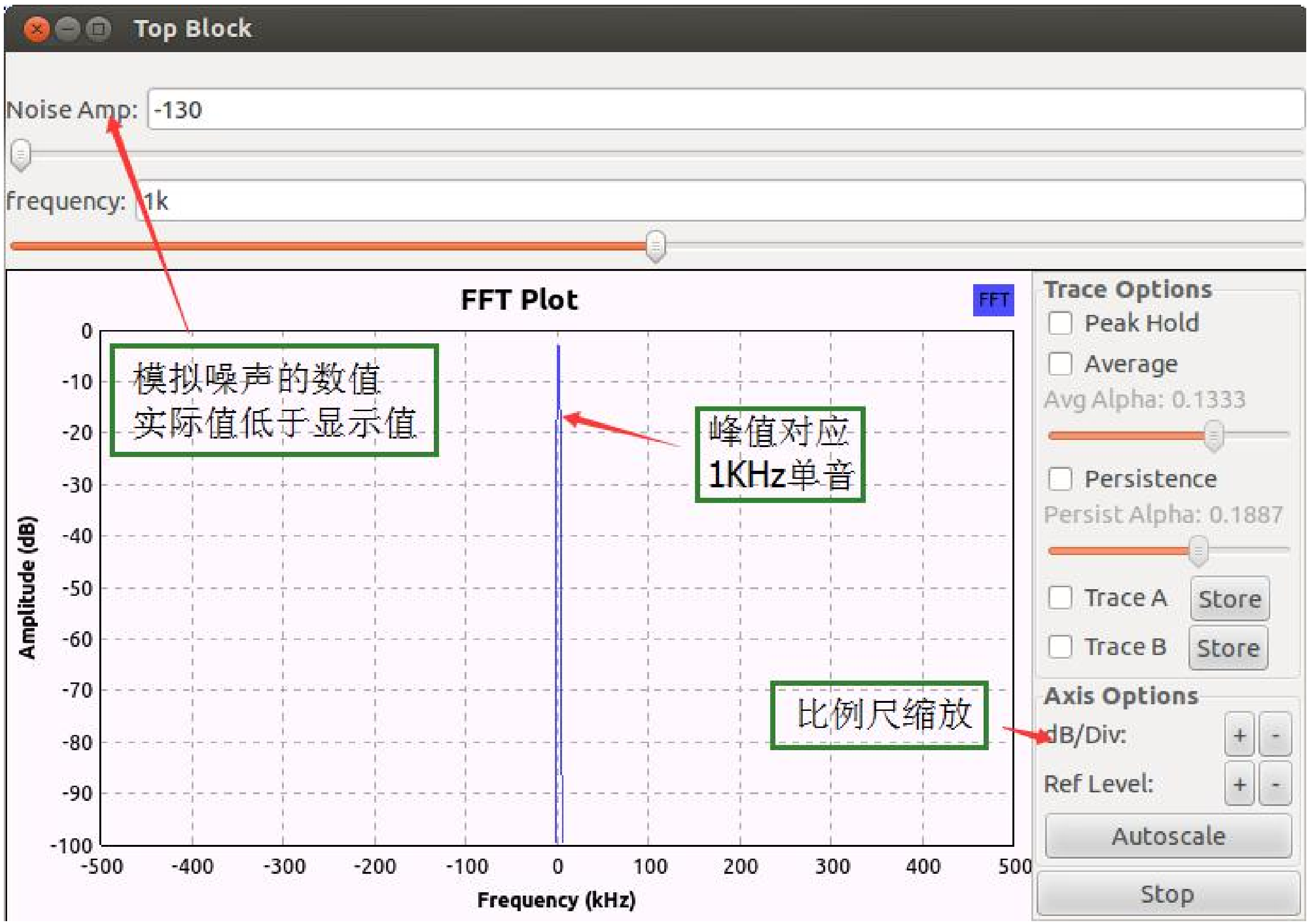


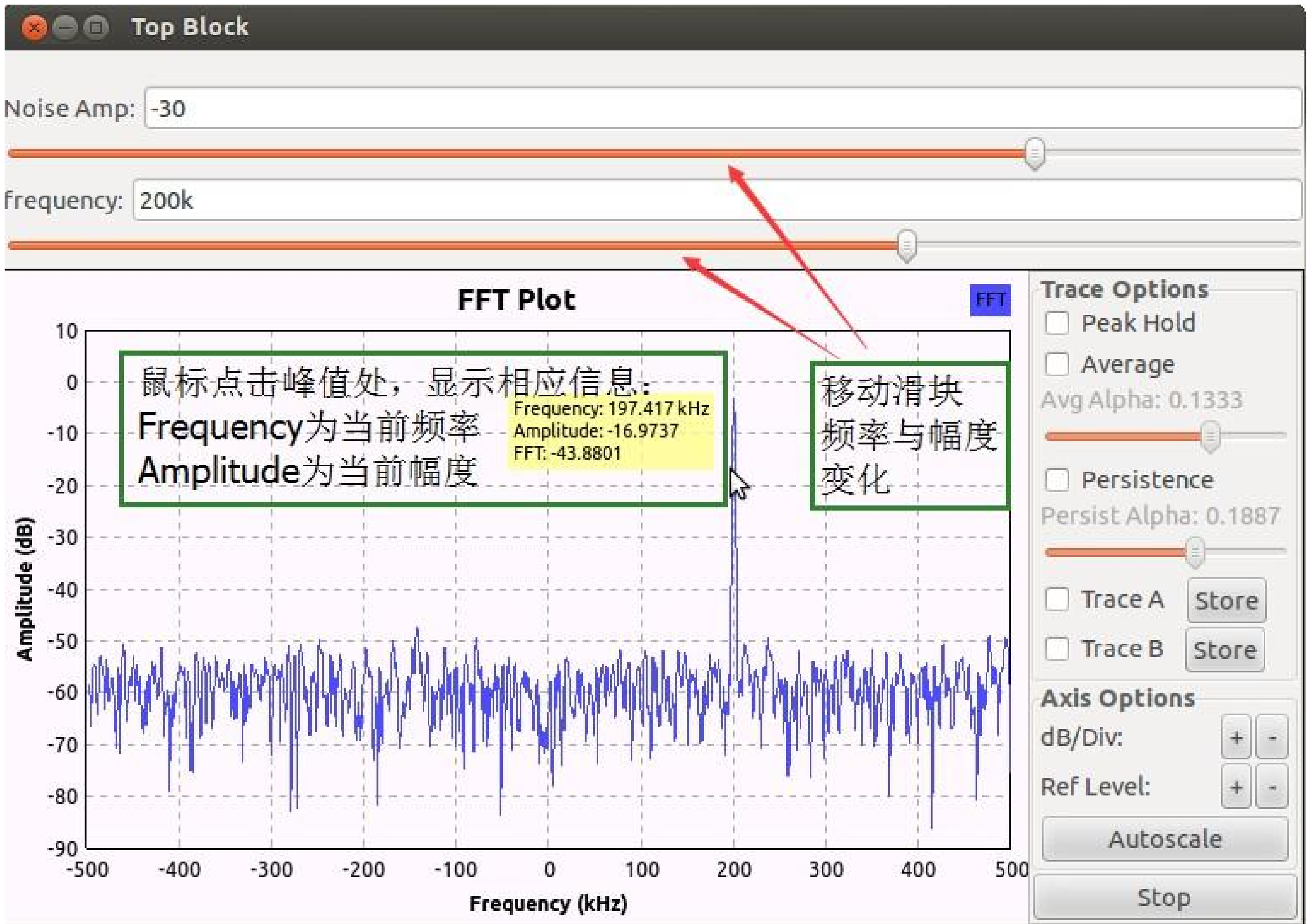

Top Block
Scope Plot
Ch1
Persistence
Analog Alpha: 0.0994
Axes Options
Secs/Div:
Counts/Div:
Y Offset:
T Offset:
Autorange
Channel Options
Ch1
Trig
Coupling:
DC
Marker:
Line Link
Stop
余弦波幅度
X轴标度
垂直偏置
水平偏置
切换自动伸缩
开关直流DC
绘制形式
Counts
Time (ms)
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
12
13
14
15
16
17
18
19
20

Lab 2：sink模块使用

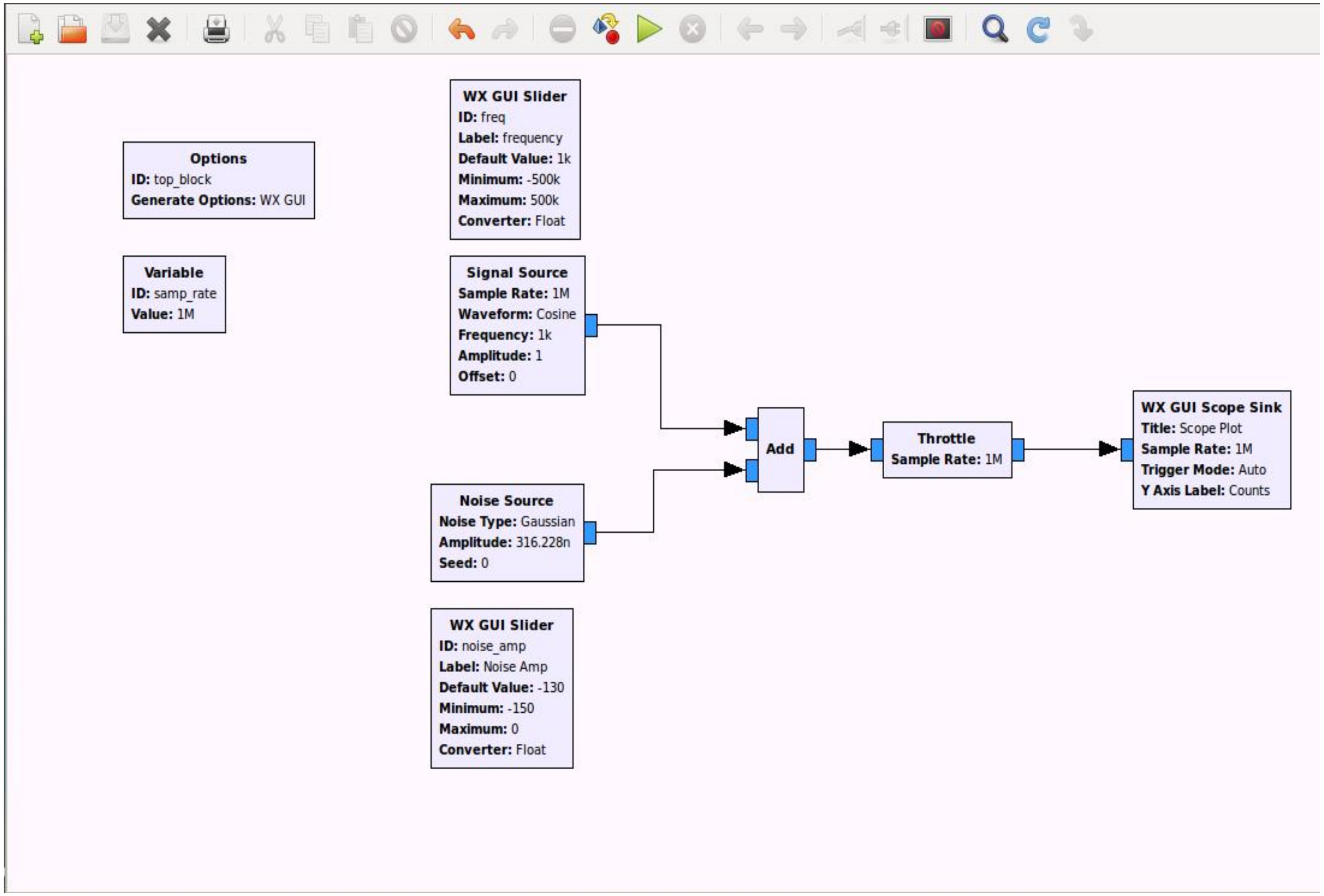



WX GUI Slider
ID: freq
Label: frequency
Default Value: 1k
Minimum: -500k
Maximum: 500k
Converter: Float
Options
ID: top_block
Generate Options: WX GUI
Variable
ID: samp_rate
Value: 1M
Signal Source
Sample Rate: 1M
Waveform: Cosine
Frequency: 1k
Amplitude: 1
Offset: 0
将数据流合到一起
Add
Throttle
Sample Rate: 1M
WX GUI FFT Sink
Title: FFT Plot
Sample Rate: 1M
Baseband Freq: 0
Y per Div: 10 dB
Y Divs: 10
Ref Level (dB): 0
Ref Scale (p2p): 2
FFT Size: 1.024k
Refresh Rate: 15
Freq Set Varname: None
Noise Source
Noise Type: Gaussian
Amplitude: 316.228n
Seed: 0
WX GUI Slider
ID: noise_amp
Label: Noise Amp
Default Value: -130
Minimum: -150
Maximum: 0
Converter: Float
噪声源模拟噪声
（滑块控制振幅）
绘制频谱图

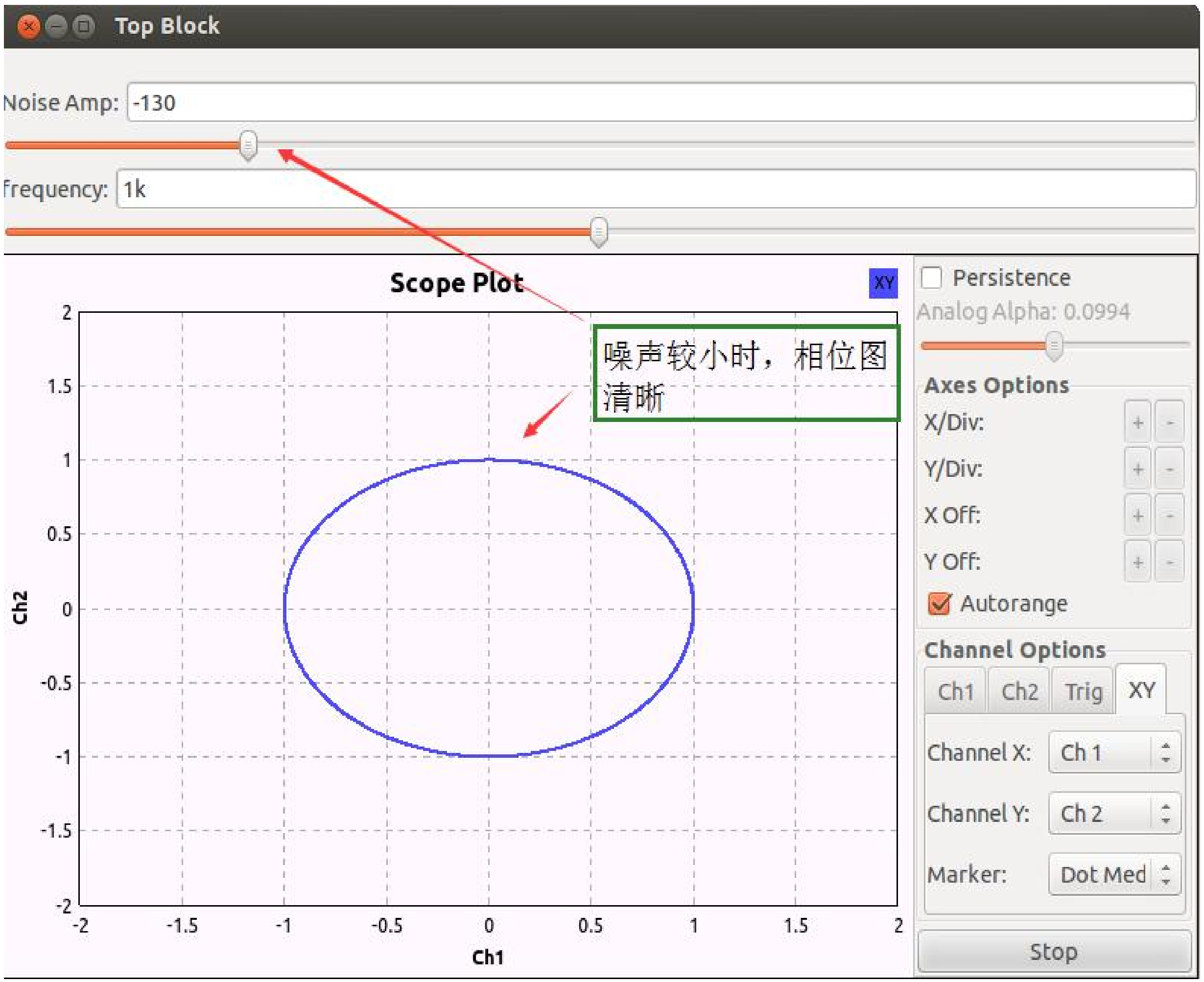

Top Block
Noise Amp: -130
frequency: 1k
FFT Plot
FFT
模拟噪声的数值
实际值低于显示值
峰值对应
1KHz单音
比例尺缩放
Amplitude (dB)
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
-500 -400 -300 -200 -100 0 100 200 300 400 500
Frequency (kHz)
Trace Options
Peak Hold
Average
Avg Alpha: 0.1333
Persistence
Persist Alpha: 0.1887
Trace A Store
Trace B Store
Axis Options
dB/Div: + -
Ref Level: + -
Autoscale
Stop



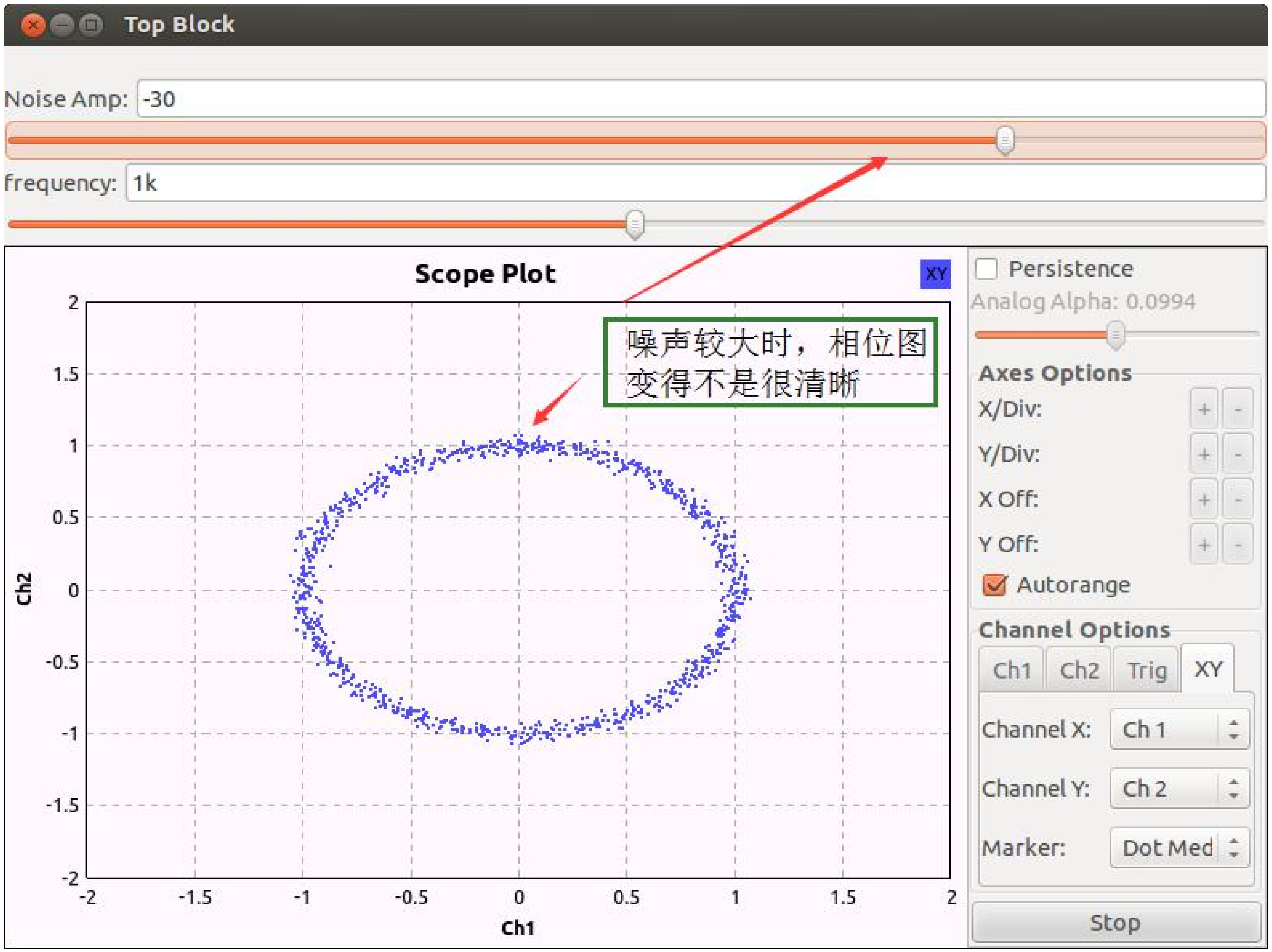
Top Block
Noise Amp: -30
frequency: 200k
FFT Plot
FFT
鼠标点击峰值处，显示相应信息：
Frequency为当前频率
Amplitude为当前幅度
Frequency: 197.417 kHz
Amplitude: -16.9737
FFT: -43.8801
移动滑块
频率与幅度
变化
Amplitude (dB)
Frequency (kHz)
10
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-500
-400
-300
-200
-100
0
100
200
300
400
500
Trace Options
Peak Hold
Average
Avg Alpha: 0.1333
Persistence
Persist Alpha: 0.1887
Trace A
Store
Trace B
Store
Axis Options
dB/Div:
Ref Level:
Autoscale
Stop



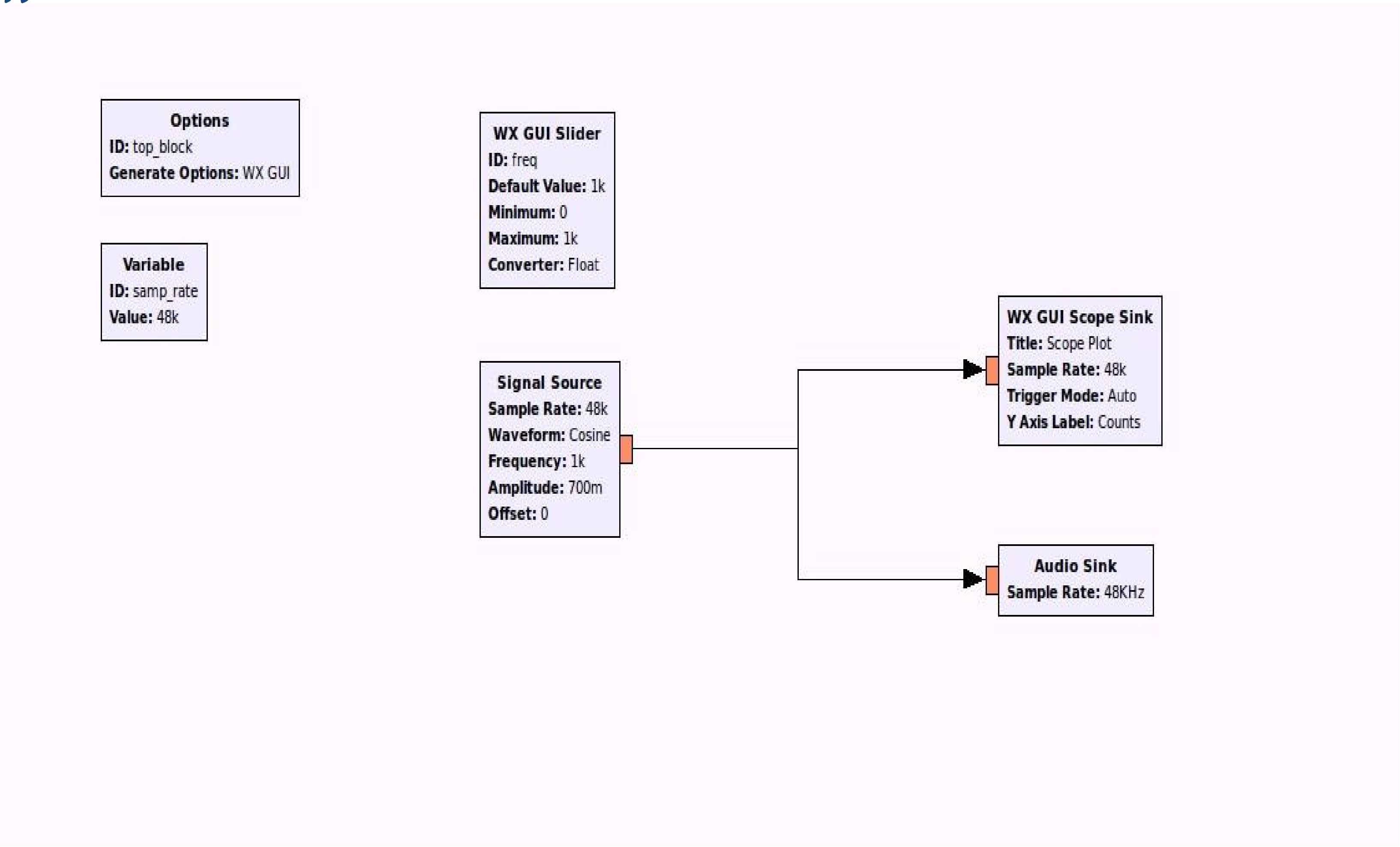
WX GUI Slider
ID: freq
Label: frequency
Default Value: 1k
Minimum: -500k
Maximum: 500k
Converter: Float
Options
ID: top_block
Generate Options: WX GUI
Variable
ID: samp_rate
Value: 1M
Signal Source
Sample Rate: 1M
Waveform: Cosine
Frequency: 1k
Amplitude: 1
Offset: 0
Add
Throttle
Sample Rate: 1M
WX GUI Scope Sink
Title: Scope Plot
Sample Rate: 1M
Trigger Mode: Auto
Y Axis Label: Counts
Noise Source
Noise Type: Gaussian
Amplitude: 316.228n
Seed: 0
WX GUI Slider
ID: noise_amp
Label: Noise Amp
Default Value: -130
Minimum: -150
Maximum: 0
Converter: Float



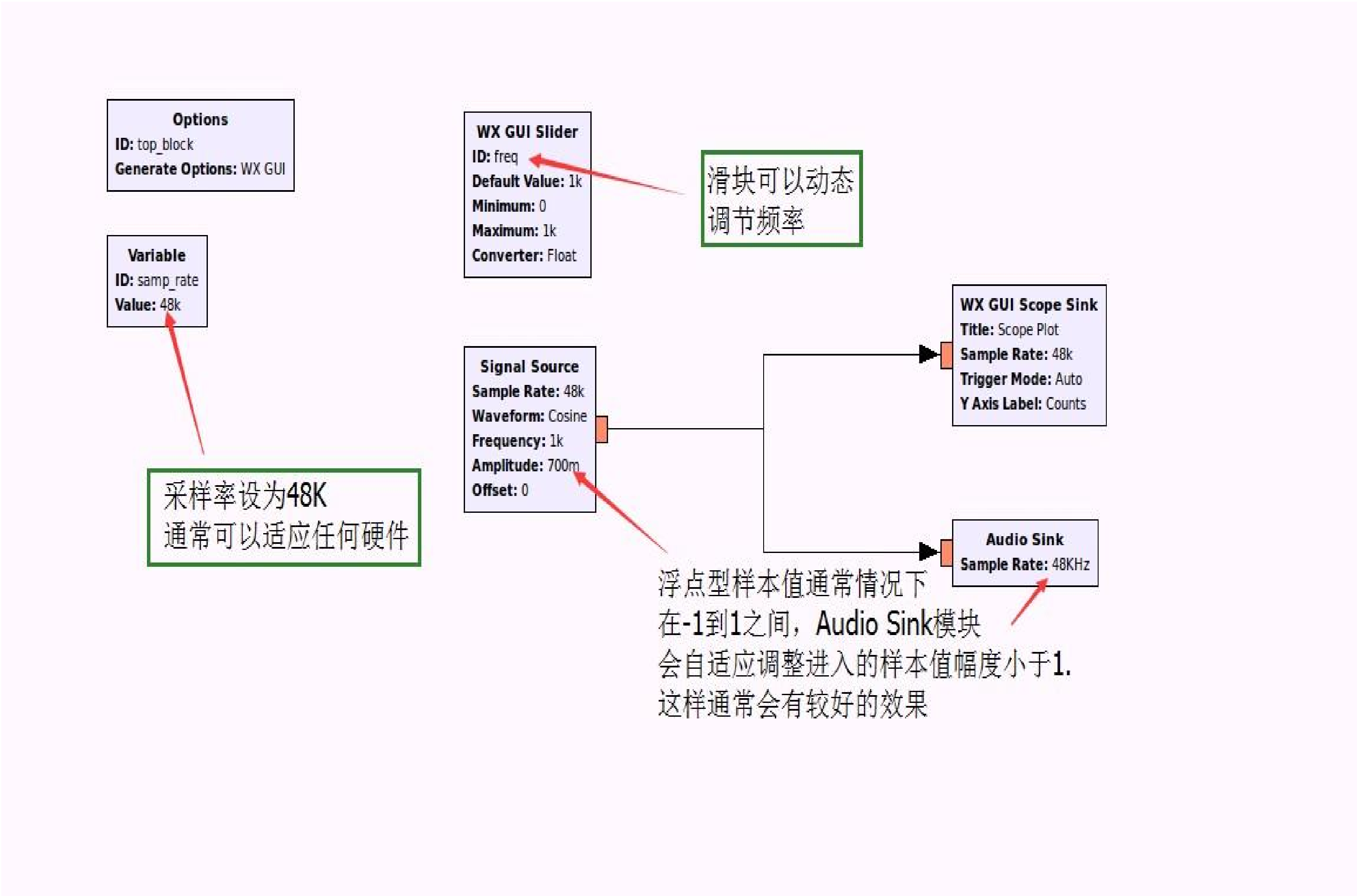
Top Block
Noise Amp: -130
Frequency: 1k
Scope Plot
XY
噪声较小时，相位图清晰
Ch2
Ch1
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
Persistence
Analog Alpha: 0.0994
Axes Options
X/Div:
Y/Div:
X Off:
Y Off:
Autorange
Channel Options
Ch1
Ch2
Trig
XY
Channel X: Ch 1
Channel Y: Ch 2
Marker: Dot Med
Stop

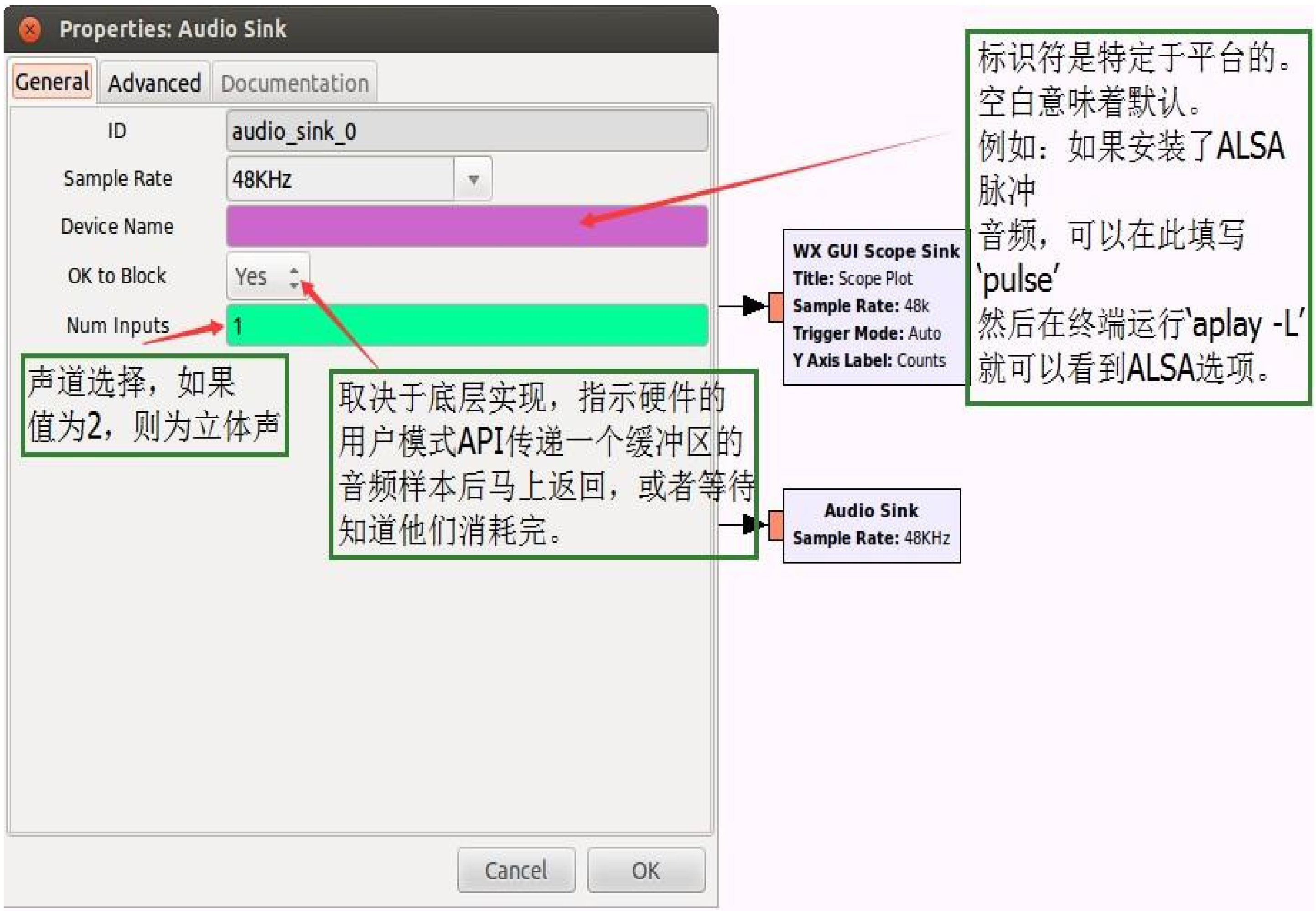

Top Block
Noise Amp: -30
frequency: 1k
Scope Plot
XY
噪声较大时，相位图变得不是很清晰
Ch2
Ch1
Persistence
Analog Alpha: 0.0994
Axes Options
X/Div:
Y/Div:
X Off:
Y Off:
Autorange
Channel Options
Ch1
Ch2
Trig
XY
Channel X: Ch 1
Channel Y: Ch 2
Marker: Dot Med
Stop

Lab 3:　音频模块使用

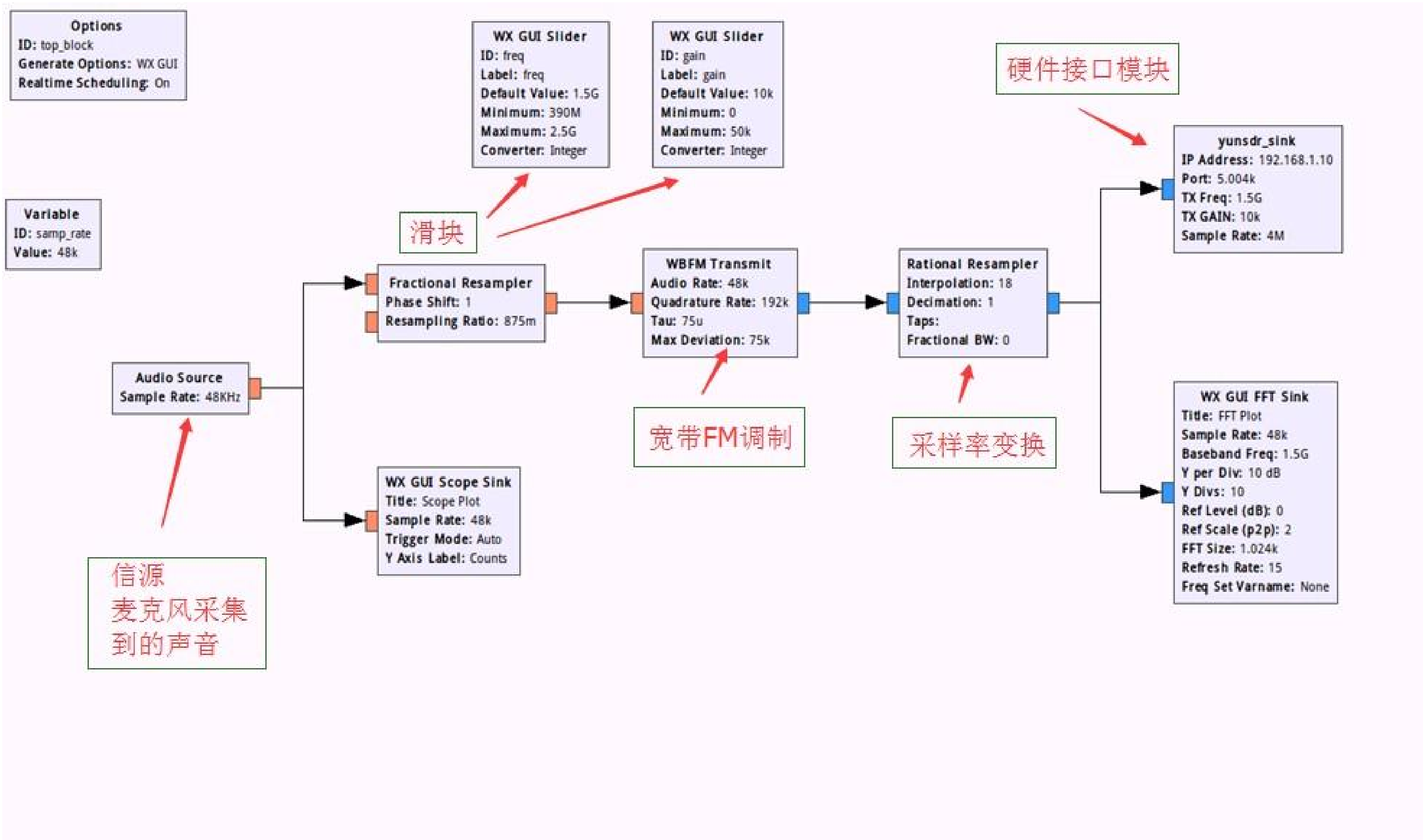


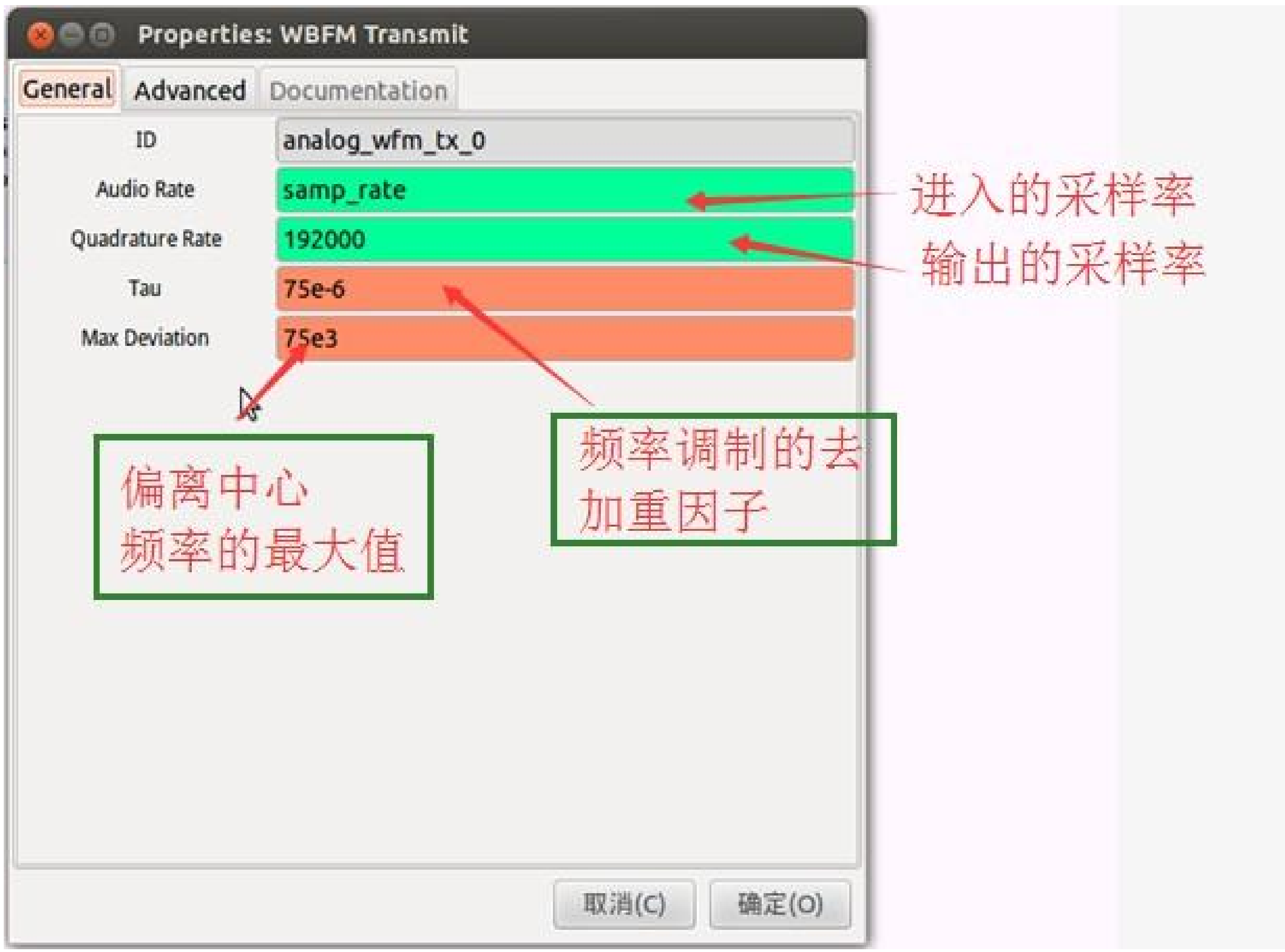
Options
ID: top_block
Generate Options: WX GUI
Variable
ID: samp_rate
Value: 48k
WX GUI Slider
ID: freq
Default Value: 1k
Minimum: 0
Maximum: 1k
Converter: Float
Signal Source
Sample Rate: 48k
Waveform: Cosine
Frequency: 1k
Amplitude: 700m
Offset: 0
WX GUI Scope Sink
Title: Scope Plot
Sample Rate: 48k
Trigger Mode: Auto
Y Axis Label: Counts
Audio Sink
Sample Rate: 48KHz

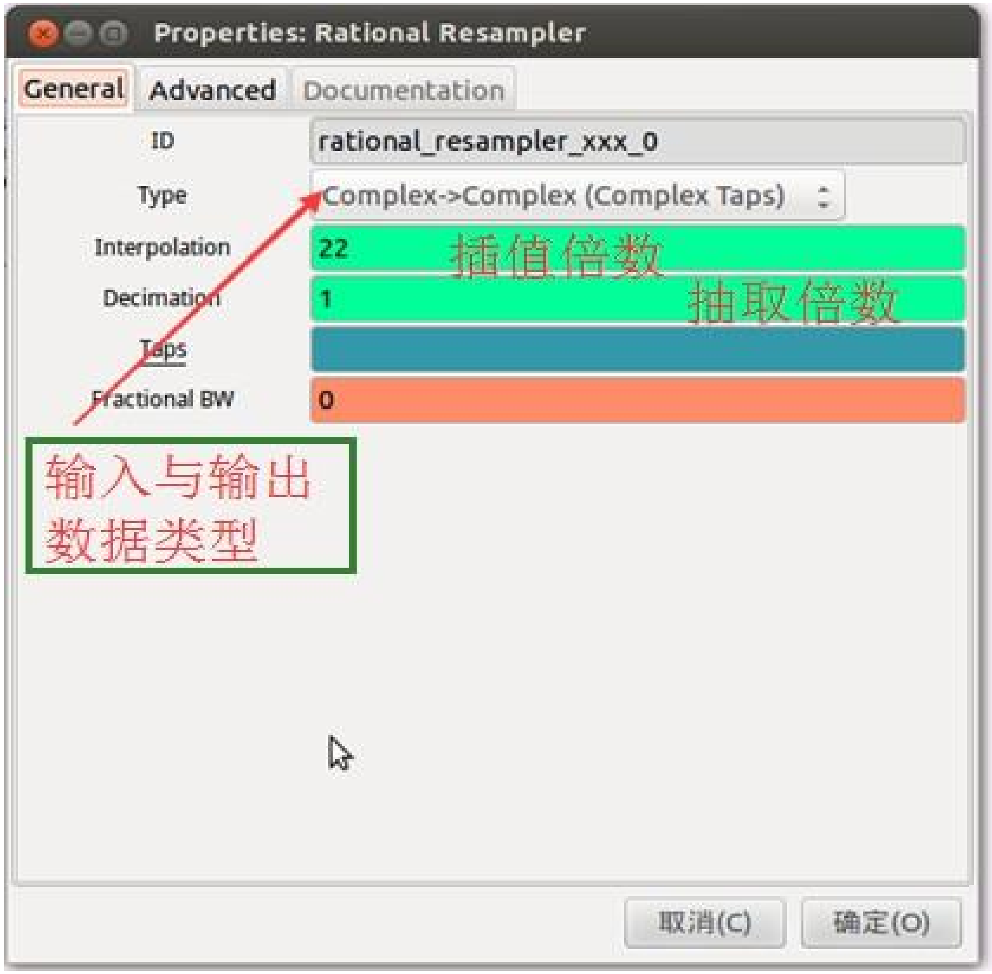

Options
ID: top_block
Generate Options: WX GUI
Variable
ID: samp_rate
Value: 48k
WX GUI Slider
ID: freq
Default Value: 1k
Minimum: 0
Maximum: 1k
Converter: Float
滑块可以动态调节频率
WX GUI Scope Sink
Title: Scope Plot
Sample Rate: 48k
Trigger Mode: Auto
Y Axis Label: Counts
Signal Source
Sample Rate: 48k
Waveform: Cosine
Frequency: 1k
Amplitude: 700m
Offset: 0
采样率设为48K
通常可以适应任何硬件
Audio Sink
Sample Rate: 48KHz
浮点型样本值通常情况下
在-1到1之间，Audio Sink模块
会自适应调整进入的样本值幅度小于1.
这样通常会有较好的效果


Properties: Audio Sink
General
Advanced
Documentation
ID
audio_sink_0
Sample Rate
48KHz
Device Name
OK to Block
Yes
Num Inputs
1
Cancel
OK
声道选择，如果值为2，则为立体声
取决于底层实现，指示硬件的用户模式API传递一个缓冲区的音频样本后马上返回，或者等待知道他们消耗完。
标识符是特定于平台的。空白意味着默认。
例如：如果安装了ALSA脉冲音频，可以在此填写‘pulse’
然后在终端运行‘aplay -L’就可以看到ALSA选项。
WX GUI Scope Sink
Title: Scope Plot
Sample Rate: 48k
Trigger Mode: Auto
Y Axis Label: Counts
Audio Sink
Sample Rate: 48KHz

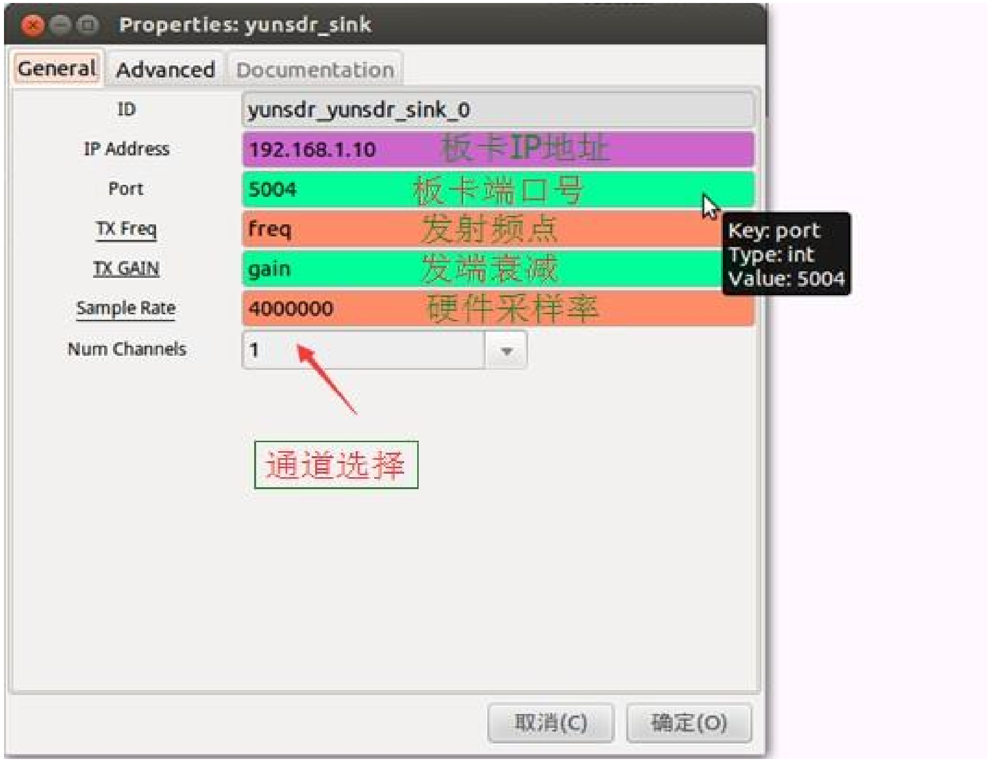
Lab 4: FM音频发射


Options
ID: top_block
Generate Options: WX GUI
Realtime Scheduling: On
WX GUI Slider
ID: freq
Label: freq
Default Value: 1.5G
Minimum: 390M
Maximum: 2.5G
Converter: Integer
WX GUI Slider
ID: gain
Label: gain
Default Value: 10k
Minimum: 0
Maximum: 50k
Converter: Integer
硬件接口模块
yunsdr_sink
IP Address: 192.168.1.10
Port: 5.004k
TX Freq: 1.5G
TX GAIN: 10k
Sample Rate: 4M
Variable
ID: samp_rate
Value: 48k
滑块
Fractional Resampler
Phase Shift: 1
Resampling Ratio: 875m
WBFM Transmit
Audio Rate: 48k
Quadrature Rate: 192k
Tau: 75u
Max Deviation: 75k
Rational Resampler
Interpolation: 18
Decimation: 1
Taps:
Fractional BW: 0
Audio Source
Sample Rate: 48KHz
宽带FM调制
采样率变换
WX GUI FFT Sink
Title: FFT Plot
Sample Rate: 48k
Baseband Freq: 1.5G
Y per Div: 10 dB
Y Divs: 10
Ref Level (dB): 0
Ref Scale (p2p): 2
FFT Size: 1.024k
Refresh Rate: 15
Freq Set Varname: None
WX GUI Scope Sink
Title: Scope Plot
Sample Rate: 48k
Trigger Mode: Auto
Y Axis Label: Counts
信源
麦克风采集
到的声音

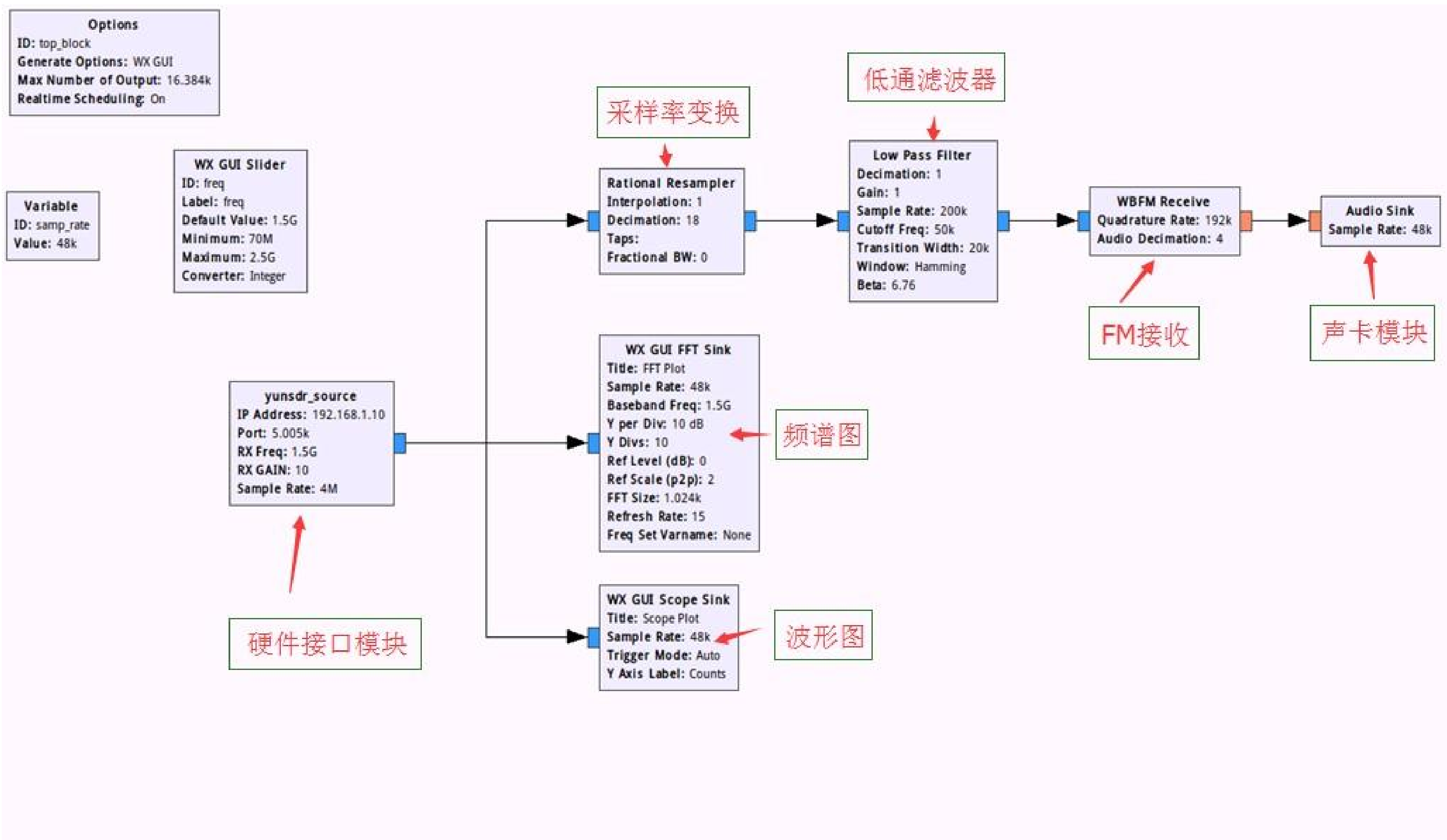

Properties: WBFM Transmit
General
Advanced
Documentation
ID
analog_wfm_tx_0
Audio Rate
samp_rate
Quadrature Rate
192000
Tau
75e-6
Max Deviation
75e3
进入的采样率
输出的采样率
偏离中心
频率的最大值
频率调制的去
加重因子
取消(C)
确定(O)



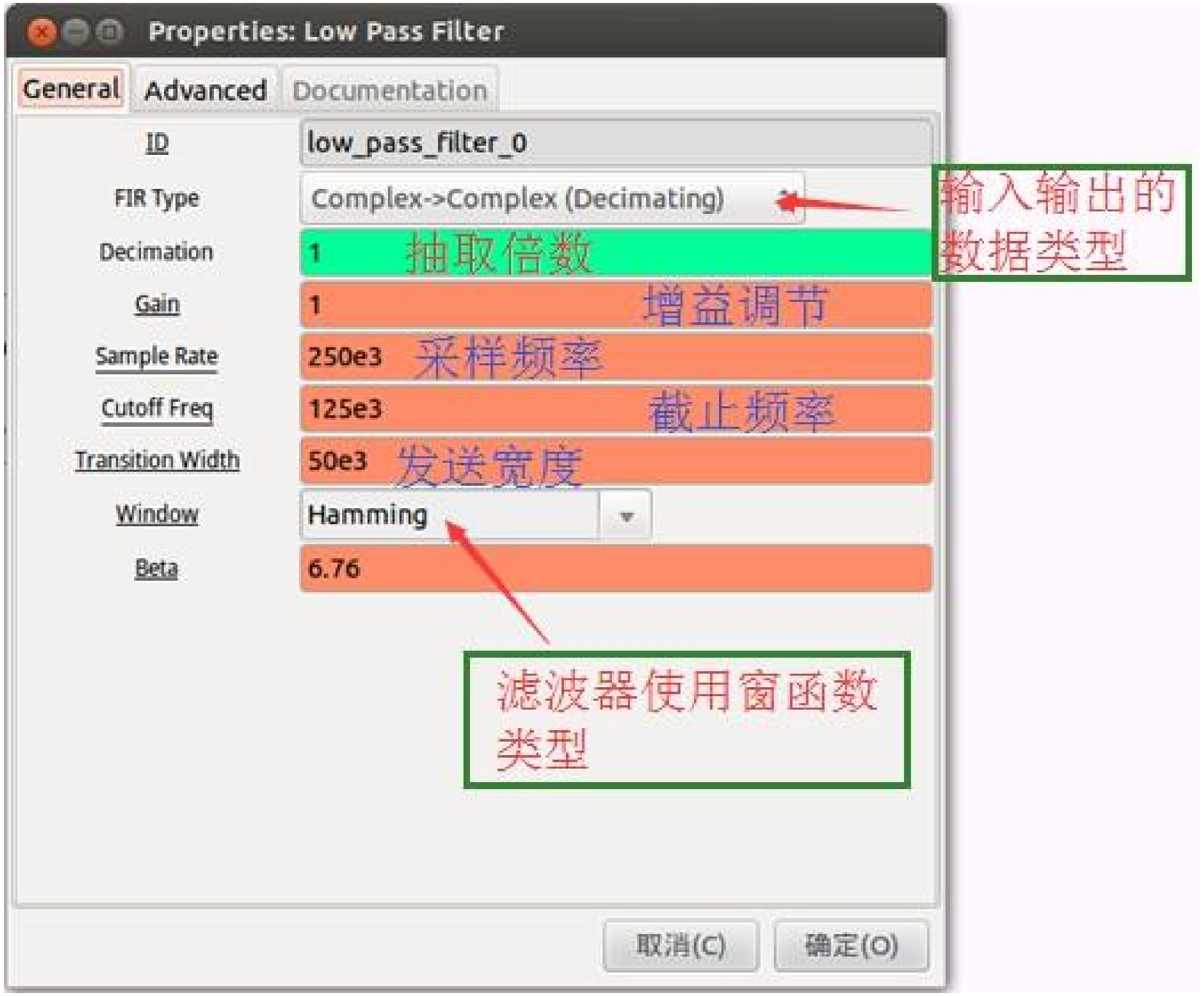
采样率变换的主要目的是调整输入采样率与输出采样率的倍数关系；往往输入采样率与输出采样率并不是整数倍关系，这就需要多采样率调节模块。

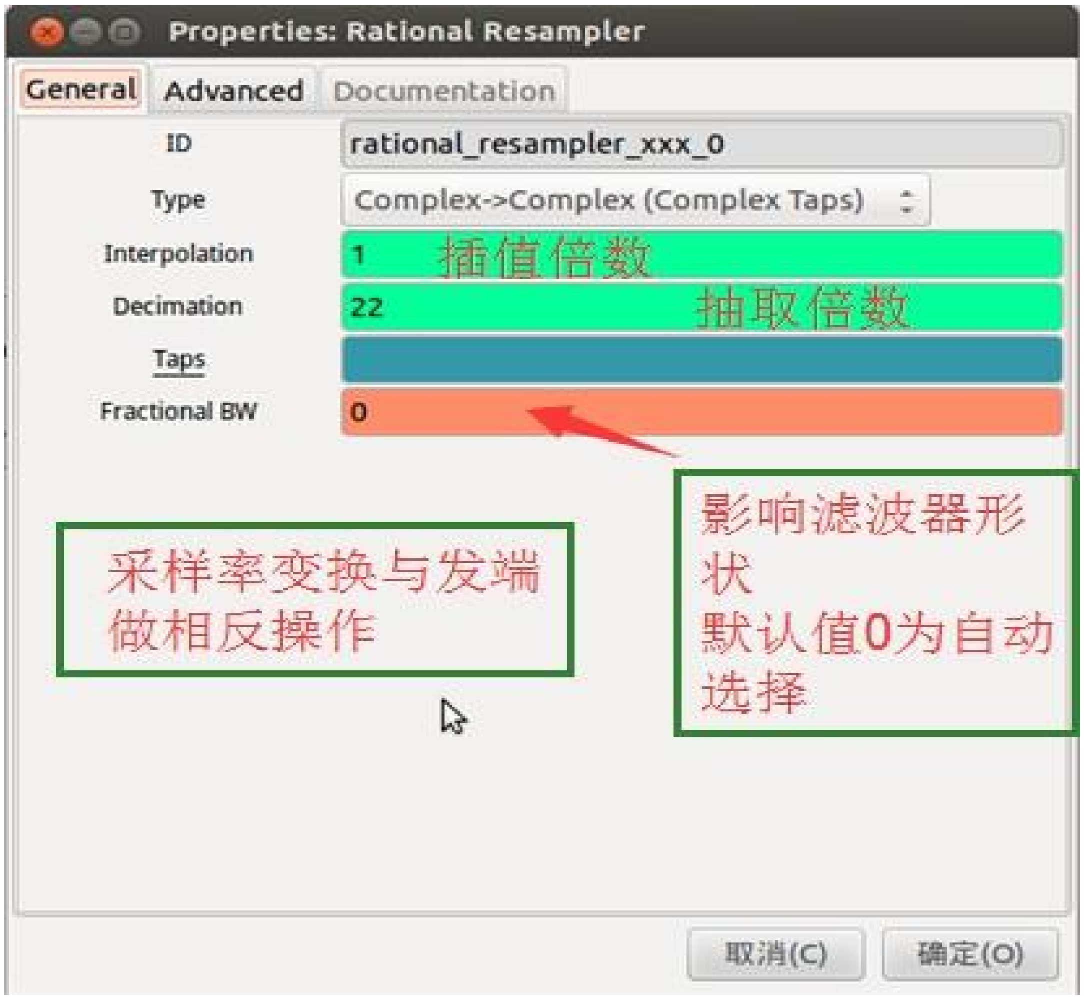

Properties: yunsdr_sink
General
Advanced
Documentation
ID
yunsdr_yunsdr_sink_0
IP Address
192.168.1.10
板卡IP地址
Port
5004
板卡端口号
TX Freq
freq
发射频点
TX GAIN
gain
发端衰减
Sample Rate
4000000
硬件采样率
Num Channels
1
Key: port
Type: int
Value: 5004
通道选择
取消(C)
确定(O)

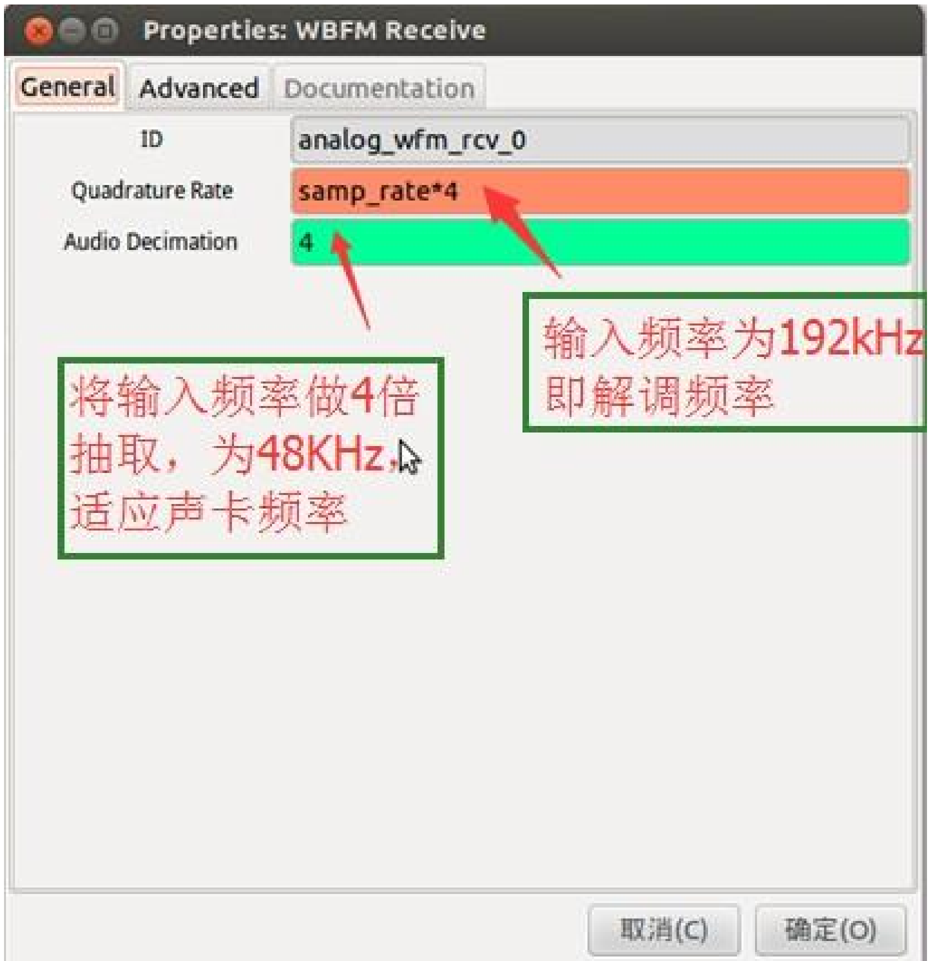
Lab 5: FM音频接收


Options
ID: top_block
Generate Options: WX GUI
Max Number of Output: 16.384k
Realtime Scheduling: On
Variable
ID: samp_rate
Value: 48k
WX GUI Slider
ID: freq
Label: freq
Default Value: 1.5G
Minimum: 70M
Maximum: 2.5G
Converter: Integer
采样率变换
Rational Resampler
Interpolation: 1
Decimation: 18
Taps:
Fractional BW: 0
低通滤波器
Low Pass Filter
Decimation: 1
Gain: 1
Sample Rate: 200k
Cutoff Freq: 50k
Transition Width: 20k
Window: Hamming
Beta: 6.76
WBFM Receive
Quadrature Rate: 192k
Audio Decimation: 4
FM接收
Audio Sink
Sample Rate: 48k
声卡模块
yunsdr_source
IP Address: 192.168.1.10
Port: 5.005k
RX Freq: 1.5G
RX GAIN: 10
Sample Rate: 4M
硬件接口模块
WX GUI FFT Sink
Title: FFT Plot
Sample Rate: 48k
Baseband Freq: 1.5G
Y per Div: 10 dB
Y Divs: 10
Ref Level (dB): 0
Ref Scale (p2p): 2
FFT Size: 1.024k
Refresh Rate: 15
Freq Set Varname: None
频谱图
WX GUI Scope Sink
Title: Scope Plot
Sample Rate: 48k
Trigger Mode: Auto
Y Axis Label: Counts
波形图


Properties: yunsdr_source
General
Advanced
Documentation
ID
yunsdr_yunsdr_source_0
IP Address
192.168.1.10
Port
5005
RX Freq
freq
RX GAIN
10
Sample Rate
4000000
收端硬件接口模块与发端参数基本相同，只是端口和增益值有所区别
取消(C)
确定(O)


Properties: Low Pass Filter
General
Advanced
Documentation
ID
low_pass_filter_0
FIR Type
Complex->Complex (Decimating)
输入输出的数据类型
Decimation
1
抽取倍数
Gain
1
增益调节
Sample Rate
250e3
采样频率
Cutoff Freq
125e3
截止频率
Transition Width
50e3
发送宽度
Window
Hamming
Beta
6.76
滤波器使用窗函数类型
取消(C)
确定(O)


Properties: Rational Resampler
General
Advanced
Documentation
ID
rational_resampler_xxx_0
Type
Complex->Complex (Complex Taps)
Interpolation
1
插值倍数
Decimation
22
抽取倍数
Taps
Fractional BW
0
采样率变换与发端做相反操作
影响滤波器形状
默认值0为自动选择
取消(C)
确定(O)


Properties: WBFM Receive
General
Advanced
Documentation
ID
analog_wfm_rcv_0
Quadrature Rate
samp_rate*4
Audio Decimation
4
输入频率为192kHz
即解调频率
将输入频率做4倍
抽取，为48KHz，
适应声卡频率
取消(C)
确定(O)